Green Industrial Transformation in Manufacturing MSMEs in Umeå:

A Co-evolutionary Analysis of Firm-level Agency and Institutional Capacity

Aaron Kondoro

Master's Thesis

Master's Programme in Spatial Planning and Sustainability

Department of Geography, Umeå University

Supervisor: Dr. Dorothee Bohn

2026

Abstract

This thesis examines how green industrial transformation among manufacturing micro, small and medium-sized enterprises (MSMEs) in Umeå, Sweden, co-evolves with local institutional capacity. Most existing work on Sweden's green transition has focused on large flagship projects, including the Northvolt gigafactory whose collapse in 2025 exposed the limits of that model. Yet around 99% of European firms are MSMEs, and the durability of the transition depends on whether institutional support actually reaches them. The study is a qualitative single-case design drawing on eight semi-structured interviews two with Umeå Municipality, five with manufacturing MSMEs, and one with a large anchor-firm manufacturer analysed through joint thematic coding alongside documentary material. The framework integrates Evolutionary Economic Geography, Regional Innovation Systems, and sustainability transitions with Benner's (2026) cultural-toolkit perspective, refined to describe co-evolution as a process rather than only its static dimensions.

Five themes emerge: the dual face of municipal action; the multi-scalar geography at which firm-side co-evolution actually happens; its asymmetric direction; the cultural register as the hinge through which institutional support is recognised or missed; and procurement, finance and binding regional conditions as the friction surfaces where the process advances or stalls. Tracing five phase-by-phase case sequences, the thesis finds that co-evolution between firms and institutions in mid-sized cities is multi-scalar, asymmetric, and culturally mediated, running only partly through the local scale. Its contribution is twofold: empirically, it documents how institutional support reaches, and largely fails to reach, the manufacturing MSME tier in one northern city after the collapse of the flagship model, with the firms' binding institutional relationships running predominantly through national, EU and

customer scales rather than the municipal one; theoretically, it relocates the cultural register to the hinge between institutional and firm sides and names the condition of latent fragmentation, in which a dense supply-side architecture coexists with uneven practical access across the firm band.

Keywords: green industrial transformation; manufacturing MSMEs; co-evolution; institutional capacity; regional innovation systems; sustainability transitions; cultural toolkit; Umeå; Sweden; mid-sized European cities.

Acknowledgements

My warmest thanks go to my supervisor, Dr. Dorothee Bohn, whose careful reading and well-timed questions strengthened this thesis.

I am grateful to the municipal officials, manufacturers, and anchor-firm representative who agreed to be interviewed, for the time and candour on which this study rests.

I also thank the Department of Geography at Umeå University, and my family and friends for their patience. Any errors are mine alone.

Table of Contents

Abstract i
Acknowledgements iii
1. **Introduction** 1
1.1 Background 1
1.2 Research Problem 5
1.3 Aim and Objectives 7
1.4 Research Questions 7
1.5 Scope and Limitations 8
1.6 Structure of the Thesis 8
2. Theoretical Framework 9
2.1 Evolutionary Economic Geography 9
2.2 Path Dependence and Path Creation 10
2.3 Co-evolution of Firms and Institutions - Concepts for a Process Account 11
2.4 Regional Innovation Systems and Institutional Capacity 13
Working definition of institutional capacity 14
2.5 Sustainability Transitions 15
2.6 The Cultural Toolkit 16
2.7 Anchor Firms and the Supplier-Tier Channel 16
2.8 Integrated Conceptual Framework 18
Figure 2.1. Integrated conceptual framework: co-evolution between MSMEs and institutions in mid-sized cities. 21
3. **Methodology** 22
3.1 Research Design and Philosophy 22
3.2 Case Selection: Why Umeå 23
3.3 Sampling and the Interview Corpus 23

Table 3.1. Interview corpus and the analytical role of each case. 25
3.4 Data Collection 26
3.5 Analytical Procedure: Joint Thematic Coding 27
Table 3.2. Consolidated thematic coding matrix across all eight interviews, including the anchor-firm case (Firm F). 33
3.6 Ethics, Anonymisation and Reflexivity 34
3.7 Quality Criteria and Limitations 34
4. Case: MSMEs and Green Transition in Umeå 36
4.1 Umeå 36
Figure 4.1. Umeå in context: from Sweden to local industrial, innovation and MSME ecosystem. 39
4.2 The Umeå Business Sector 40
Table 4.1. Distribution of Umeå employment by firm size-class, 2023. 41
4.3 Green Transition Objectives and Policy Actors 41
Figure 4.2. The Umeå innovation ecosystem for manufacturing MSMEs. 45
5. **Findings and Discussion** 46
5.1 Reading All Data Sources Together 46
5.2 Institutional Supply and the Dual Face of Municipal Action 47
5.3 The Multi-Scalar Geography of Co-evolution 49
5.4 Asymmetric Co-evolution: Who Adapts to Whom 53
5.5 Cultural Toolkits as the Interpretive Hinge of Co-evolution 56
5.6 Procurement and Finance as Cross-Cutting Friction Surfaces 59
5.7 The Process of Co-evolution: Five Case Sequences 62
6. **Conclusions** 67
6.1 Answers to the Research Questions 67
Main research question 67
Sub-question 1 68
Sub-question 2 68
Sub-question 3 69
6.2 Empirical and Theoretical Contributions 70
6.3 Policy Implications 71

6.4 Limitations and Future Research.....72
References.....74

1. Introduction

1.1 Background

Much of the conversation about Sweden's green industrial transformation has tended to centre on a relatively small set of high-profile places. The Northvolt gigafactory in Skellefteå, the Hybrit fossil-free steel pilot, and the H2 Green Steel/Stegra plant in Boden have absorbed the press attention, the venture capital, and most of the descriptive language used to talk about the Swedish green model abroad (OECD, 2023; OECD, 2025a). Then in March 2025, Northvolt filed for bankruptcy. The picture changed.

It changed in a way that should concern people far outside Sweden. Bloomberg's review at the end of 2024 found that twelve of sixteen planned European-led battery factories had already been delayed or cancelled (Fortune, 2024). Britishvolt collapsed into administration in 2023; the German cell-maker CustomCells followed in 2025; ACC halted construction of two gigafactories in Germany and Italy in 2024; FREYR Battery in Norway paused cell manufacturing later that year (Eurofound, 2025; S&P Global Mobility, 2025). The flagship-led model large, capital-intensive, state-backed mega-projects designed to carry whole sectors at once is in trouble across Europe (Tagliapietra & Trasi, 2024).

My argument in this thesis is that something more durable will have to be built on a broader and less visible base: the micro, small and medium-sized manufacturers that actually make up most of any country's industrial fabric, and the municipal institutions that surround them. The flagship is not where the test is. The test is in places like Umeå.

Why should anyone outside Sweden care about how a few hundred manufacturing MSMEs in a mid-sized northern city handle decarbonisation? Three reasons. First,

Sweden is the OECD's stress test. Carbon pricing covers more than 95% of fossil emissions, the legally binding net-zero target is 2045, and supporting institutions (Industrial Leap, Climate Leap, the 23 sectoral roadmaps under Fossil Free Sweden) are among the most generous anywhere in Europe (OECD, 2025a; Karltorp & Maltais, 2024). If green industrial transformation cannot find traction in this institutional environment, it is unlikely to find it elsewhere on the continent.

Second, the post-flagship lesson scales. The Inflation Reduction Act in the United States, the EU's Clean Industrial Deal of February 2025, and the gigafactory programmes in Germany, France, Spain and the United Kingdom have all bet heavily on a similar large-project playbook (European Commission, 2025a). The recent setbacks suggest that this playbook needs a complement, not a replacement. Third, Umeå is not exotic. Hundreds of mid-sized cities across Europe Trondheim, Oulu, Tampere, Aberdeen, Bilbao, Brno, Linz, Reggio Emilia share its rough profile: a few large anchor firms, a manufacturing MSME base, an ambitious municipal climate commitment, and a regional innovation system that is strong in some places and thin in others. What works or fails in Umeå is, in that sense, a template question for that whole category of place.

The European policy backdrop has shifted enough in the last two years that this argument is sharper now than when the research was first designed. The Net-Zero Industry Act and Critical Raw Materials Act, both adopted in 2024, gave the Green Deal teeth (European Commission, 2020; Florence School of Regulation, 2024). The Clean Industrial Deal of February 2025 added something the earlier instruments largely lacked: an explicit promise to draw small and medium-sized enterprises into sectoral transition pathways rather than leaving them to absorb regulation drafted with large firms in mind. Most recently, Jolly et al. (2025) have argued that the next research agenda for green regional industrial path development needs to ask

explicitly who benefits, at what scale, and on whose terms a critical question this thesis takes seriously.

A note on terms. I treat green industrial transformation, drawing on Trippl, Baumgartinger-Seiringer, Frangenheim, Isaksen and Rypestøl (2020) and Markard, Raven and Truffer (2012), as the process through which firms, institutions, and regional systems reshape their technologies, production routines, organisational practices, and governance arrangements toward ecological sustainability in response to regulatory pressure, market change, and broader societal expectations, and in ways conditioned by place-specific historical and institutional conditions.

I use the term MSMEs micro, small and medium-sized enterprises rather than the more common SMEs. The European Commission's Recommendation 2003/361/EC distinguishes three size categories within the MSME band: micro-enterprises (under 10 employees), small enterprises (10-49), and medium-sized enterprises (50-249), with a combined annual turnover ceiling of fifty million euros (European Commission, 2003). Strict SME analyses sometimes drop the micro tier. In the Umeå case that would be a serious analytical loss: micro-enterprises are the most numerous category of local firms and a substantial share of local manufacturing employment (Frisk & Fineman, 2025). Within that population, I distinguish between two analytically different kinds of MSME. Incumbent firms are established manufacturers, often family-owned, often with several decades of regional history. Green start-ups are recently founded firms whose primary activity is environmental technology, circular production, or sustainability-oriented services. The two face different pressures and draw on different cultural repertoires when responding to them (Hug, Mayer & Seidl, 2024; Benner, 2026).

MSMEs make up over 99% of all firms in the EU and employ roughly two thirds of the private-sector workforce (European Commission, 2023). Yet the literature on

green industrial transformation has been heavily skewed toward big actors: gigafactories, national strategies, high-visibility clusters (Hug et al., 2024; Madrid-Guijarro & Duréndez, 2024). MSMEs face transformation pressures with thinner internal resources, less specialist expertise, and a stronger reliance on whatever institutional support their region happens to provide (Tödtling & Trippl, 2005; Yunusbayeva, Pohl, Hjelmberg & Bhatti, 2025).

Cultural framing matters too. Benner (2026) argues that regions and firms move in particular directions partly because they share repertoires of stories, identities, and ideas about what counts as a sensible response to a problem. These cultural toolkits do not determine outcomes, but they shape what local actors regard as plausible, legitimate, or worth trying. In northern European contexts where regional identities around forestry, craftsmanship, and self-reliance are still very much alive this dimension is hard to ignore.

Northern Sweden is where most of the action is concentrated geographically. Norrbotten and Västerbotten together are expecting industrial investments on the order of EUR 110 billion over the next two decades (Nordregio, 2024; OECD, 2023). The Swedish government's Strategy for New Industrialisation and Societal Transformation in Norrbotten and Västerbotten, launched in June 2024, sets out seven target areas covering electricity infrastructure, housing, schooling, and skills (Government of Sweden, 2024). Umeå the largest urban centre in the north and the administrative seat of Västerbotten County is part of this story without being its loudest voice. It is one of seven Swedish cities selected for the EU Mission for 100 Climate-Neutral and Smart Cities by 2030 (Swedish Energy Agency, 2022), it has signed and renewed the Climate City Contract 2030 (Viable Cities, 2025), and it has set a long-term target of reaching 200,000 residents by 2050 within an explicit sustainability framing (Umeå Municipality, 2024).

Recent business-sector data make the case sharper. Between 2018 and 2023 Umeå's value added grew by 45%, against a Swedish national average of 37%. Employment grew by 11% against 8% nationally. The local economy now counts roughly 6,667 firms and 36,400 employees, with around 84% of local employment sitting within the MSME range (Frisk & Fineman, 2025). The detailed size-class breakdown is set out in Chapter 4. Manufacturing accounted for 35% of employment in 2023. The city hosts a small number of large anchor manufacturers in vehicle, forestry-machinery, life-science and process-equipment sectors anonymised throughout this thesis alongside a tier of smaller suppliers and processors that are exactly the MSMEs of interest here.

1.2 Research Problem

Sustainability transitions research has expanded quickly. But one question has been left mostly untouched: how do manufacturing MSMEs actually engage in green industrial transformation when the relevant institutions are at the municipal scale? Evolutionary Economic Geography (EEG) has shown that regional development tends to follow path-dependent trajectories, and that firms inherit much of their room for manoeuvre from earlier industrial structures (Martin & Sunley, 2006; Boschma & Frenken, 2006). Recent work on co-evolution has added that firms and institutions adapt in response to one another, rather than developing independently (Gong & Hassink, 2019; Chu & Hassink, 2023). Three gaps remain.

The first gap is theoretical. EEG has been criticised, with some justification, for treating space as a backdrop rather than as part of the explanation, and for not engaging strongly enough with the normative side of sustainability what should change, who decides, and on whose terms (Essletzbichler, 2012; Kogler et al., 2023). Cultural framings, regional imaginaries, and shared meaning-making have only recently begun to find their way into mainstream EEG analysis (Benner, 2026). At

the same time, the recent literature has called for a much clearer process account of how co-evolution actually unfolds: through which mechanisms, at which scales, and along which channels (Trippl et al., 2020; Frangenheim, 2023; Jolly et al., 2025). The framework developed in Chapter 2 takes this call seriously.

The second gap is empirical. Most existing work focuses on large industrial actors, flagship decarbonisation projects, or core metropolitan regions. Mid-sized cities in northern peripheries have received much less attention (Dubois & Carson, 2016; Isaksen & Trippl, 2017). Where MSMEs have been studied, the analysis often stops at the firm boundary rather than asking how those firms actually interact with the municipal planning system or the Regional Innovation System around them (Hug et al., 2024; Yunusbayeva et al., 2025).

The third gap is policy-relevant. The European Commission's 2024 Country Report for Sweden flags concerns about SMEs in lagging regions explicitly, and the Just Transition Fund channels EUR 24 million toward the metallurgical industry in Västerbotten (European Commission, 2024). But there is no clear empirical account of how MSMEs in places like Umeå actually use, ignore, or work around these institutional supports. After the Northvolt bankruptcy, the Swedish government, the Commission, and policymakers across Europe are visibly rethinking how the next phase of green industrial policy should be structured (Council of the European Union, 2025; European Commission, 2025a).

This thesis sets out to fill that gap by examining how green industrial transformation among manufacturing MSMEs in Umeå distinguishing between incumbent firms and green start-ups emerges through co-evolutionary processes shaped by path dependence, institutional capacity, local agency, and place-specific cultural resources.

1.3 Aim and Objectives

The aim of this thesis is to examine how green industrial transformation among manufacturing MSMEs in Umeå emerges through the co-evolution between firm agency and local institutional capacity. Four objectives follow. The first is to describe what manufacturing MSMEs in Umeå are actually doing in the name of green industrial transformation. The second is to analyse how local institutional conditions affect what these firms can attempt. The third is to look at the back-and-forth between MSMEs and municipal institutions, attending to both the formal mechanisms and to the cultural repertoires that mediate them (Benner, 2026; Gong & Hassink, 2019). The fourth is to draw out what this case can tell us theoretically and for policy about green industrial transformation in mid-sized northern cities.

1.4 Research Questions

Main Research Question: How does green industrial transformation among manufacturing MSMEs in Umeå co-evolve with local institutional capacity and planning systems?

Sub-question 1: How do manufacturing MSMEs in Umeå including both incumbent firms and green start-ups engage in green industrial transformation through changes in production practices, technologies, or business models?

Sub-question 2: How do local institutional conditions including planning systems, support structures, and policy frameworks influence MSMEs' ability to pursue green transformation?

Sub-question 3: How do interactions between MSMEs and municipal institutions contribute to continuity or change in local industrial development pathways?

1.5 Scope and Limitations

The scope is bounded geographically (Umeå Municipality, with Region Västerbotten included where relevant), sectorally (manufacturing MSMEs in construction-related supply chains, bio-based industries, marine and environmental technology, and process technology, together with one large anchor-firm manufacturer), and definitionally (the European Commission's Recommendation 2003/361/EC). The study is a qualitative case study based on eight semi-structured interviews and document analysis, so it does not claim statistical generalisability. The aim is analytical generalisation to comparable mid-sized cities and northern peripheries. The policy environment has been moving fast: the Clean Industrial Deal (February 2025), the December 2025 simplification deal for sustainability reporting, the Northvolt bankruptcy in March 2025, and the wider European pattern of gigafactory delays have all changed the landscape during fieldwork. Findings should be read as situated within this moving target.

1.6 Structure of the Thesis

Chapter 2 develops the theoretical framework, with explicit attention to the concepts needed to describe co-evolution as a process. Chapter 3 sets out the methodology, including the joint coding procedure through which all data sources municipal interviews, firm interviews, the anchor-firm interview, and documents were analysed together. Chapter 4 maps the case empirically. Chapter 5 reports and discusses the findings together, organised thematically around the five common themes that emerged from joint coding, then traces co-evolution as a process through five case sequences. Chapter 6 concludes by answering the research questions directly, stating the empirical, theoretical and policy contributions, and identifying limitations and directions for future research. The next chapter develops the analytical architecture needed to interpret the empirical material.

2. Theoretical Framework

This chapter develops the analytical framework needed to interpret the empirical material in Chapter 5. It draws together four bodies of work: Evolutionary Economic Geography (EEG), with its emphasis on path dependence and path creation; the literature on co-evolution between firms and institutions; Regional Innovation Systems (RIS) theory; and sustainability transitions research, particularly the Multi-Level Perspective (MLP). These four are then read through Benner's (2026) cultural toolkit perspective, which gives the framework a way to handle the meaning-making side of green industrial transformation. A central analytical aim of the chapter is to refine the framework so that it can describe co-evolution as a process not only its dimensions. That refinement is the work of section 2.3, which sets out the specific concepts the analysis will use to track the process, and of section 2.7, which names the anchor-firm channel as one route through which firm-level transformation enters the local economy. The integrated architecture is presented as Figure 2.1 at the end of the chapter.

2.1 Evolutionary Economic Geography

Evolutionary Economic Geography (EEG) emerged in the early 2000s as an attempt to bring evolutionary thinking from economics into the geographical analysis of regional development (Boschma & Frenken, 2006; Boschma & Martin, 2010). Its starting claim is that regional economies do not optimise from scratch in response to changing conditions. They evolve, in the sense that the routines, competences, and firm populations of any given region carry forward into the next period, shaping what is possible and what is not. EEG treats regional development as a path-dependent process (Martin & Sunley, 2006), embedded in particular industrial structures (Frenken, Van Oort & Verburg, 2007), and shaped by the specific knowledge bases that firms in the region have built over time (Asheim & Gertler,

2005). Geography is not a setting in this account; it is a constitutive part of the explanation (Chu & Hassink, 2023; Kogler et al., 2023). Dubois and Carson (2016) develop this point further for the north Swedish context, arguing that northern regions need to be understood through their specific economic-geographical conditions rather than as deviations from a metropolitan default.

EEG has been productively criticised. Essletzbichler (2012) argued that early EEG work tended to underplay institutions and political economy. More recent work has pulled the framework toward a richer engagement with institutions, agency, and meaning-making (Grillitsch & Sotarauta, 2020; Benner, 2022, 2026; Kogler et al., 2023). It is in this enriched form that EEG enters the present thesis: as a framework that takes regional history seriously without treating it as deterministic, and that is increasingly comfortable with the idea that institutional and cultural factors do real causal work.

2.2 Path Dependence and Path Creation

Path dependence is the EEG concept that has had the longest reach. The basic idea, developed in the economics literature by David (1985) and applied to regions by Martin and Sunley (2006), is that small early choices can lock a region into a particular development trajectory through self-reinforcing mechanisms increasing returns, accumulated investments, network effects, and the gradual build-up of competences and institutions around a particular industry. Once a path is established, it becomes hard to change, even when external conditions shift in ways that would in principle favour a different trajectory.

Path dependence on its own is, however, an incomplete account of regional change, because it has difficulty explaining how new paths emerge. This is where Garud and Karnøe's (2001) concept of path creation becomes central. Their argument, extended

in subsequent work (Garud, Kumaraswamy & Karnøe, 2010), is that new development paths typically emerge through what they call mindful deviation: the deliberate mobilisation of existing resources by actors who are trying to do something different. Path creation is not a clean break with the past. It is a strategic recombination of inherited assets in a way that opens up new possibilities.

Trippl, Baumgartinger-Seiringer, Frangenheim, Isaksen and Rypestøl (2020) take this argument further for the specifically green case. They show that green regional industrial path development depends on three interacting elements: regional preconditions (existing industrial structures, organisational support, institutional set-ups and natural-resource endowments), asset modification (the modes through which those preconditions are actively reshaped), and agency (the actors who do the reshaping). Frangenheim (2023) extends the framework outward, showing that the preconditions enabling a path in one region are increasingly shaped by what happens in other regions path development is interpath as well as intra-regional. Jolly et al. (2025) sharpen the critical edge of this literature, asking who benefits from green path development, at what scale, and on whose terms. Taken together, these three pieces give this thesis the vocabulary it needs to describe Umeå's green industrial transformation as path creation built on path-dependent foundations not as a clean break from northern industrial history, but as a deliberate recombination of inherited assets across local, national and EU scales.

2.3 Co-evolution of Firms and Institutions - Concepts for a Process Account

If path dependence and path creation describe how regional trajectories unfold over time, the concept of co-evolution describes how firms and institutions shape one another within those trajectories. Gong and Hassink (2019) provide the most explicit recent treatment, defining co-evolution in EEG as a process in which firms and the institutions around them adapt in mutual response neither developing independently

of the other. The framework distinguishes between several directions of co-evolution: firms acting on institutions, institutions acting on firms, and mutual adaptation in which the influence runs both ways simultaneously.

To describe the process of co-evolution not only its dimensions the analysis below leans on five specific concepts. The first is direction: in any concrete instance, who is adapting to whom, and how reciprocally? Gong and Hassink (2019) and Chu and Hassink (2023) insist that co-evolution can be one-way, two-way, or asymmetric, and that the analytical work is to specify which. The second is scale: at what spatial level is the adaptation actually happening local, regional, national, or supranational? Frangenheim (2023) shows that the relevant scale is rarely confined to a single region, and Coenen, Benneworth and Truffer (2012) make the same point for sustainability transitions. The third is channel: through which concrete mechanism does the mutual adaptation pass? Channels in this thesis include municipal land allocation, the Industriråd dialogue, the Eco Industrial Park, public procurement rules, EKN export-finance guarantees, EU innovation grants, the incubator system, and customer sustainability-reporting demands. The fourth is asset modification the specific reshaping of regional preconditions through which co-evolution materially occurs (Trippl et al., 2020). The fifth is the cultural register through which actors recognise themselves as participating in the same project, or fail to (Benner, 2026; Swidler, 1986). The analytical chapters use these five concepts to describe how co-evolution actually unfolds in the Umeå case.

By “institutions” I follow the influential formulation developed by Hodgson (2006), who defines institutions as the systems of established and prevalent social rules that structure social interaction. On this definition institutions encompass both formal arrangements laws, regulations, planning systems, and the public agencies that enact them and informal arrangements such as routines, conventions, and shared norms

(North, 1990; Hodgson, 2006). The classical distinction between institutions (the rules of the game) and organisations (the players acting under those rules) follows North (1990), but for the analytical purposes of this thesis I treat the boundary as porous, since municipal bodies such as Umeå Municipality embody and produce institutional rules at the same time as they act under them.

Benner's more recent work (2022, 2026) extends this in two directions. First, it argues that industrial-institutional coevolution is multidimensional rather than singular: institutions do not act on firms through a single channel but through several simultaneously regulatory, organisational, network-based, and cultural. Second, it argues that co-evolution operates not only through formal institutional channels but also through cultural ones, in which shared interpretive resources allow firms and institutions to recognise themselves as participating in the same project (Benner, 2026). This second move is what allows the cultural toolkit perspective developed in section 2.6 to be situated within the broader co-evolutionary framework rather than functioning as an exogenous addition to it.

2.4 Regional Innovation Systems and Institutional Capacity

The Regional Innovation Systems (RIS) tradition developed parallel to EEG and overlapping with it. Originating in the work of Cooke (2001) and Asheim and Gertler (2005), RIS treats regions as configurations of innovation-supporting actors and institutions: universities, research centres, public agencies, intermediaries, financial institutions, and firms themselves. The strength of a region's innovation capacity, in this tradition, depends on the density of these actors and the quality of the connections between them.

Tödtling and Trippl (2005) developed a typology that has become canonical: regions can be organisationally thin (lacking key support actors), fragmented (having actors

but lacking the connections between them), or locked in to mature industrial paths in ways that block transformation. The typology has been extended in more recent work, notably by Isaksen, Trippl and Mayer (2022), who distinguish between regions that are merely reorienting (adjusting at the margins toward sustainability) and regions that are pursuing transformation (actively redirecting their industrial trajectory). The distinction is empirically important: as Chapter 5 will show, Umeå reads more like a transformation-oriented RIS than like a thin or fragmented one. But Doloreux and Parto's (2005) early warning still applies institutional thickness is not the same as institutional accessibility, and dense networks can produce coordination costs that fall hardest on smaller firms.

Coenen, Asheim, Bugge and Herstad (2017) push the RIS tradition closer to the EEG one by arguing that contemporary regional innovation systems are evolving toward what they call mission-orientation: a deliberate broadening of the system's purpose, beyond competitiveness alone, toward addressing societal challenges including climate change. This shift is consequential for the analysis here, because it means that municipal action toward green industrial transformation is no longer a side activity within a competitiveness-oriented innovation system; it is increasingly the system's main purpose.

Working definition of institutional capacity

Drawing the discussion of EEG, co-evolution and RIS together, this thesis adopts the following working definition of institutional capacity, which guides the empirical analysis in Chapter 5. Institutional capacity refers to a region's combined ability to (i) configure its formal organisational architecture the public agencies, universities, intermediaries, support programmes, and planning systems within which firms operate; (ii) maintain accessible network connections between those organisations and the firms they are designed to serve; (iii) sustain shared cultural-

interpretive resources that allow institutions and firms to recognise themselves as participating in the same project (Benner, 2026); and (iv) adjust over time through co-evolutionary feedback from firm experience back to institutional action (Gong & Hassink, 2019). This definition draws together strands from Hodgson (2006), Tödtling and Trippl (2005), Isaksen, Trippl and Mayer (2022), Coenen et al. (2017), Benner (2022, 2026), and Gong and Hassink (2019).

2.5 Sustainability Transitions

The sustainability transitions literature provides the third leg of the framework. Its central conceptual device is the Multi-Level Perspective (MLP), developed by Geels (2002) and extended in subsequent work (Geels, 2019; Markard, Raven & Truffer, 2012). The MLP analyses sociotechnical change at three interacting levels: the niche, where new technologies and practices are protected and developed; the regime, which is the dominant socio-technical configuration in a given sector; and the landscape, the slow-moving external context of climate change, demographic shifts, and macro-economic patterns within which niche and regime interact.

The niche concept is particularly relevant for the empirical analysis. Strategic Niche Management theory (Kemp, Schot & Hoogma, 1998; Smith, Voß & Grin, 2010) argues that protected experimental spaces are necessary for green technologies to mature against an incumbent regime that would otherwise outcompete them on cost, scale, or familiarity. The Eco Industrial Park described in Chapter 5 functions as this kind of planned niche space. Coenen, Benneworth and Truffer (2012) added an explicitly spatial dimension to this debate, arguing that transitions are place-specific in ways the MLP literature initially underplayed.

2.6 The Cultural Toolkit

Benner's (2026) cultural toolkit perspective is the integrating layer of the framework. The argument is straightforward but consequential. Regions and firms move in particular directions partly because they share repertoires of stories, identities, and ideas about what counts as a sensible response to a problem. These cultural toolkits do not determine outcomes, but they shape what local actors regard as plausible, legitimate, or worth trying. The concept builds on Swidler's (1986) original sociological formulation of culture as a toolkit of available strategies of action, but extends it to the regional and institutional level in ways that fit naturally with EEG's spatial concerns.

For the present thesis, the cultural toolkit concept does two specific kinds of analytical work. First, it gives a way to read Umeå's municipal framing of itself as a "living circular hub" as something more than rhetoric as an institutional imaginary that selects which firms are recognised as part of the municipal project. Second, it gives a way to read each firm's own register the substantive vocabulary the interviewee uses to describe what sustainability means in their business as part of the co-evolutionary process rather than as background colour. When the registers align, co-evolution gains a cultural grip; when they diverge, the channels still operate but the recognition that makes them feel relevant is absent.

2.7 Anchor Firms and the Supplier-Tier Channel

One concept central to co-evolutionary dynamics in mid-sized industrial regions, but not yet fully integrated into the framework above, is the anchor firm. Anchor firms belong in a co-evolutionary account of the manufacturing tier because the channels through which institutional capacity reaches smaller firms do not all run directly. A large incumbent manufacturer can transmit sustainability requirements, technical standards and agglomeration externalities down its supplier tier, making it a

structurally distinct conduit of co-evolution that a firm tier composed only of independent MSMEs would miss. Setting the concept out here, alongside the other theoretical building blocks, allows the framework to specify this conduit in advance, so that the cases and the data can later be read against it. The literature on anchor firms in regional development now spans more than two decades from the original anchor-tenant hypothesis (Agrawal & Cockburn, 2003; Feldman, 2003) through its extension to thin regional innovation systems (Karlsen & Lund Pedersen, 2013) and its more recent productivity-effects formulation in the Swedish context (Nilsson, Schubert & Miörner, 2025). The brief synthesis offered here picks up the strands most relevant to a co-evolutionary reading and connects them to the co-evolutionary process concepts developed in section 2.3.

An anchor firm, for the purposes of this thesis, is a large manufacturing establishment whose presence in a region structures the local supplier ecosystem and innovation environment around it (Agrawal & Cockburn, 2003; Karlsen & Lund Pedersen, 2013; Nilsson et al., 2025). The defining features are four. First, scale: an anchor firm is well above the MSME ceiling of 250 employees and has the procurement volume to support a tier of smaller suppliers (European Commission, 2003). Second, embeddedness: it has been in the region long enough for those supplier relationships to have stabilised into routines, technical specifications, and audit conventions that smaller firms then carry forward as path-dependent assets (Boschma & Frenken, 2006; Martin & Sunley, 2006). Third, externality generation: large, locally established firms generate agglomeration externalities demand thickening, knowledge spillovers, labour-pooling, and reputational signals that benefit smaller firms in the surrounding cluster (Agrawal & Cockburn, 2003; Feldman, 2003; Nilsson et al., 2025). Nilsson, Schubert and Miörner's (2025) recent panel-data analysis of Swedish regions between 2007 and 2019 shows that these

productivity effects are real, statistically robust, and disproportionately important for firms in smaller, non-core regions which makes the anchor-firm concept especially relevant to mid-sized northern cities such as Umeå. Fourth, signal-setting in sustainability terms: an anchor firm's own technical, environmental and sustainability requirements propagate down the supplier tier whether or not those requirements take a formal contractual form (Karltorp & Maltais, 2024; Trippl et al., 2020), and recent value-chain evidence indicates that large in-scope corporates are increasingly using supplier-tier engagement as the primary mechanism through which sustainability obligations reach the MSME band (European Commission, 2025a).

2.8 Integrated Conceptual Framework

The conceptual framework developed across section 2.1-2.7 is visualised in Figure 2.1 as a layered architecture. At the top, landscape pressures the EU Green Deal, the Clean Industrial Deal, carbon pricing, the CSRD/CSDDD reporting regime, and supply-chain sustainability demands act on three interlocking theoretical columns: EEG (path dependence and path creation), RIS (institutional capacity), and Sustainability Transitions/MLP (niche-regime-landscape). Beneath the theoretical columns sits the institutional tier, organised explicitly into four spatial scales municipal, regional (RIS), national, and EU/supranational reflecting the multi-scalar reading of co-evolution developed in section 2.3 (Coenen et al., 2012; Frangenheim, 2023; Isaksen, Trippl & Mayer, 2022).

The horizontal band running between the institutional and firm tiers represents the cultural register the interpretive hinge through which actors on both sides recognise (or fail to recognise) themselves as participating in the same co-evolutionary project. Locating Benner's (2026) cultural-toolkit perspective at this hinge position, rather than placing it within either tier, captures the analytical work the perspective is

meant to do: it operates between formal supply and firm-side experience. Channels of co-evolution planning and land allocation, procurement, regulatory approval, sustainability reporting, customer demand, export finance, and the supplier-tier channel run vertically through the hinge and connect specific institutional locations to specific firm-side units.

The firm tier itself is organised into three elements rather than two, reflecting the elaboration of section 2.7. The first is incumbent MSMEs: established, often family-owned manufacturers carrying path-dependent assets and adapting incrementally under regulatory and market pressure. The second is green start-up MSMEs: sustainability-embedded or green-niche firms whose products and identities are oriented toward the green transition from founding or near-founding, and whose agency runs more toward path creation. The third is the anchor firm and its supplier tier, the structurally distinct element introduced in section 2.7 (Agrawal & Cockburn, 2003; Karlsen & Lund Pedersen, 2013; Nilsson, Schubert & Miörner, 2025). Distinguishing three firm-side elements completes the architecture: the framework can specify, for each element, which channels of co-evolution are active, at which scale, and in which direction. The supplier-tier channel running between the anchor firm and the surrounding MSME suppliers is shown as a distinct channel, since it is the conduit through which a large incumbent's sustainability requirements and technical standards can propagate into the MSME band without passing through the municipal interface.

Two further conventions complete the architecture. Solid grey arrows running from the institutional tier through the hinge to the firm tier represent the institution-to-firm direction of co-evolution. Dashed gold arrows running upward through the hinge represent the firm-to-institution feedback channel: the cultural-toolkit feedback through which firm-level agency reshapes institutional action (Benner,

2026). The line connecting the anchor-firm element to the municipal institutional tier represents the supplier-tier-and-anchor channel, a bidirectional channel through which anchor-firm requirements can reach the surrounding MSME suppliers. The process concepts of section 2.3 direction, scale, channel, asset modification, and cultural register are noted at the foot of the figure as the analytical vocabulary through which any specific co-evolutionary instance is described.

The integrated framework is what makes the empirical material in Chapter 5 readable as evidence about how green industrial transformation actually unfolds. Each empirical claim in the findings can be located within the architecture by specifying which channel is active, at which scale, in which direction, through which firm-side element, and with what kind of cultural-register alignment. The same chapter then uses this architecture to trace co-evolution as a process, not only as a set of static dimensions.

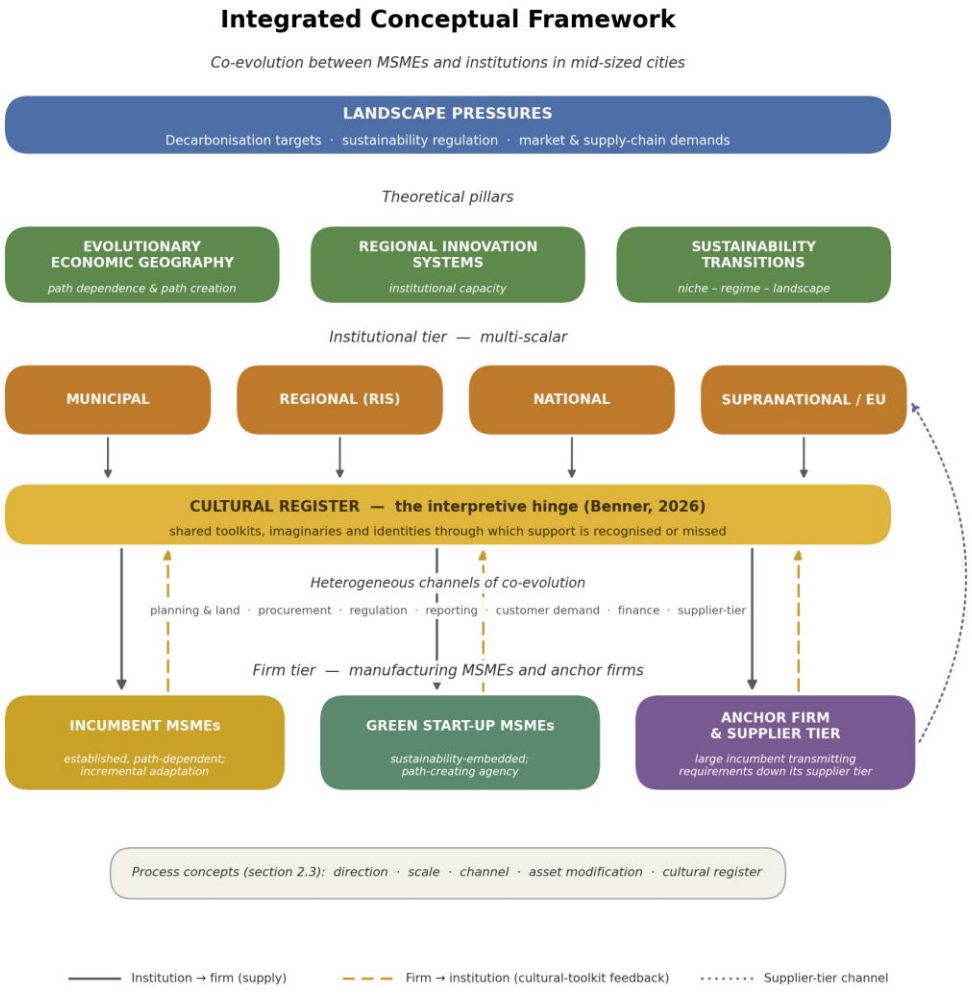


Figure 2.1. Integrated conceptual framework: co-evolution between MSMEs and institutions in mid-sized cities. The framework integrates EEG, RIS and Sustainability Transitions under landscape pressures; locates the cultural register as a horizontal hinge between institutional and firm tiers (Benner, 2026); and organises the firm tier into three elements incumbent MSMEs, green start-up MSMEs, and the anchor-firm-and-supplier-tier connected to the institutional tier through heterogeneous channels of co-evolution. Source: author, drawing on Boschma and Frenken (2006), Trippl et al. (2020), Frangenheim (2023), Gong and Hassink (2019), Tödtling and Trippl (2005), Coenen et al. (2012), Geels (2002), Benner (2026), Agrawal and Cockburn (2003), Karlsen and Lund Pedersen (2013), Nilsson, Schubert and Miörner (2025), and Jolly et al. (2025).

3. Methodology

This chapter explains how the thesis was carried out. It sets out the research design, the case-selection logic, the sampling strategy, the data-collection procedures, the thematic analytical approach (revised in this version to code all data sources together rather than separately), the ethical considerations including the anonymisation protocol, and the limitations of the design. Each choice is connected back to a specific element earlier in the thesis: the research questions in Chapter 1, the conceptual architecture in Chapter 2, and the case description in Chapter 4.

3.1 Research Design and Philosophy

The study uses a qualitative, single-case design with embedded units. The single case is Umeå Municipality and its manufacturing MSME ecosystem. The embedded units are eight interviews: two with Umeå Municipality officials, five with manufacturing MSMEs, and one with the communications director of an incumbent anchor-firm manufacturer (as defined in section 2.7). This design suits the research questions, all three of which ask how something happens. “How” questions are well-suited to in-depth case work where the goal is to understand process and meaning rather than to estimate prevalence (Yin, 2018; Stake, 1995). They are less well-suited to large-N statistical work, which would lose the relational dimensions the framework treats as central.

Philosophically, the study sits within a critical realist register. It accepts that institutional and industrial structures are real and have causal effects, while recognising that those structures are accessed through actors’ interpretations of them (Bhaskar, 2008; Sayer, 2000; Yeung, 2019). Critical realism is a productive home for EEG work because it lets the analysis hold path dependence, institutional

configurations, and cultural meaning-making in the same frame without collapsing one into the other.

3.2 Case Selection: Why Umeå

The methodological case for Umeå rests on three criteria. First, theoretical relevance: Umeå has a sizeable and growing manufacturing MSME base (Frisk & Fineman, 2025), an explicit climate commitment under the Climate City Contract 2030 (Viable Cities, 2025), and a regional innovation system rated as a leader but unevenly distributed (Region Västerbotten, 2022). The case is theoretically rich rather than typical (Flyvbjerg, 2006). Second, empirical accessibility: the fieldwork was carried out while the researcher was based at Umeå University. Third, contemporary salience: Northvolt's bankruptcy in March 2025, the Clean Industrial Deal in February 2025, and the December 2025 simplification deal for sustainability reporting (Council of the European Union, 2025) have all changed the policy environment during fieldwork.

3.3 Sampling and the Interview Corpus

The sampling logic is purposive and theoretically driven. Two principles guided the construction of the corpus. The first is that institutional capacity must be observed from both the supply side (municipal officials describing what is offered) and the demand side (MSMEs describing what they experience), to avoid the supply-side bias the literature has identified (Doloreux & Parto, 2005; Tödtling & Trippl, 2005). The second is the incumbent / green start-up distinction set out in Chapter 1 and represented as the two-lane MSME tier of Figure 2.1.

All MSME firms and individual interviewees are anonymised throughout the thesis. The five completed MSME interviews are referred to as Firm A, Firm B, Firm C, Firm D, and Firm E. Their sectoral profiles, size bands, and analytical roles are

described in sufficient detail for the analysis to be intelligible without compromising anonymity. Municipal officials are identified by role only (Business Director and Business Developer), since the positions are public and unique within Umeå Municipality.

Interview	Actor type	Role / firm profile (anonymised)	Analytical role
Interview 1	Municipal - Strategic	Head of Economic Development (Business Director), Umeå Municipality	Strategic-brokerage view; first-contact and connector role; selective firm-attraction logic.
Interview 2	Municipal - Operational	Business Developer / Näringslivsutvecklare, Umeå Municipality	Operational view; Umeå Eco Industrial Park; project-based pipelines; soft industrial governance.
Interview 3	Firm A - Green start-up (Micro)	Owner-founder of a micro-firm in biological agri-tech; biodegradable carrier substitution	Customer-led transformation; weak municipal contact; national regulatory bottleneck (Naturvårdsverket). Owner-translator dyad.
Interview 4	Firm B - Listed software group	Investor-relations lead of a multi-unit listed software group; headquartered in Umeå with ~80 employees in Umeå.	Group-level sustainability through governance, reporting and customer products; few Umeå-based customers; tests how MSME-style transformation looks for a firm locally embedded as employer but not as customer-served firm.
Interview 5	Firm C - Owner-led manufacturing (Small)	Owner-CEO and Operations Lead of a small manufacturer; ~12 employees	Owner-led path creation through solar, EVs, geothermal, green aluminium, tree-planting; internally financed and ethically motivated; partial use of

Interview	Actor type	Role / firm profile (anonymised)	Analytical role
			Klimatklivet/Naturvårdsverket; politically distant.
Interview 6	Firm D - Export-oriented marine technology (Small/Medium)	CEO of a green marine-technology firm headquartered in Umeå; ~15 employees; exports to over 100 countries.	Sustainability-embedded export firm; long-life non-chain mooring + floating-solar enabling tech; EU grant support; municipality has "zero" direct role; national export-finance gap is the binding constraint.
Interview 7	Firm E - Globally connected micro-firm	Business Manager of a micro-firm with seven employees in advanced process technology; operates in over 40 countries.	Sustainability-embedded process technology; RIS-embedded through incubator, Vinnova, EU programmes and Business Sweden; conservatism in regulated sectors is the main adoption barrier.
Interview 8	Firm F - Anchor-firm manufacturer	Communications director of a large incumbent vehicle-cab and pressed-steel plant in Umeå; ~1,800 employees; nearly a century of regional history; part of a global manufacturing group.	Anchor-firm perspective; tests the supplier-tier channel (see section 2.7), the Industriråd municipal interface, multi-scalar drivers, and the regional enabling conditions (infrastructure, electricity, competence).

Table 3.1. Interview corpus and the analytical role of each case. All firms and individual interviewees are anonymised. Source: author.

Three observations follow. First, the municipal interviews are not redundant. Interview 1 captures the strategic, brokerage-oriented role; Interview 2 captures the operational, project-based role. Reading them together produces the two-faced institutional architecture discussed in section 5.2. Second, the five MSME interviews

are deliberately diverse rather than directly comparable. Third, the anchor-firm interview (Firm F) occupies the analytically distinct slot defined in section 2.7: it is the only case in the corpus that can speak to the supplier-tier channel and to the municipal Industriråd interface from the side of a large incumbent manufacturer.

3.4 Data Collection

All eight interviews were semi-structured. Three interview guides were prepared one for municipal officials, one for MSMEs, and one for the anchor firm all following the same logic: open questions about the actor's role, structured questions about specific themes drawn from the framework, and probing questions about gaps that emerged during the conversation (Brinkmann & Kvale, 2018; Galletti, 2013). Interviews ranged from approximately 35 to 75 minutes. Seven of the eight interviews were conducted remotely via Microsoft Teams: both municipal interviews, four of the five MSME interviews (Firms A, B, D and E), and the anchor-firm interview (Firm F). The Firm C interview was conducted by telephone. The Teams interviews were captured as video recordings; Firm A was captured in audio-only mode on Teams at the participant's preference. The audio track was extracted from each video file before transcription. All recordings were transcribed verbatim and entered into the joint coding system described in section 3.5. The Firm A interview was mediated by the owner's child as translator; the Firm C interview was conducted in two parts, the second a short follow-up with the operations lead reflecting on the firm's bicycle programme for employees. The anchor-firm interview was conducted with the plant's communications director, which shapes the kind of evidence it provides: the interview gives a strong account of the plant's strategic position, municipal relationship and binding regional conditions, while detailed supplier-tier questions are directed by the respondent to logistics,

purchasing and engineering functions a boundary that is itself an analytically meaningful finding (see section 5.7).

Remote interviewing via Microsoft Teams has a small but real methodological footprint. On the strengths side, video-mediated interviewing allowed scheduling flexibility for time-constrained MSME respondents, reduced the observer effect that physical presence in a firm's premises can produce, and made consistent recording straightforward (Archibald, Ambagtsheer, Casey, & Lawless, 2019). On the limits side, it constrained access to the non-verbal cues, premises, and material context that in-person fieldwork would have made visible a limitation managed through documentary triangulation rather than eliminated.

The interview corpus is supported by documentary material: the Dun & Bradstreet Näringslivsanalys for Umeå (Frisk & Fineman, 2025) and Västerbotten County (Fineman & Vieweg, 2026); the Umeå Eco Industrial Park presentation materials shared by the Business Developer during Interview 2; Umeå Municipality publications; the Climate City Contract 2030 (Viable Cities, 2025); the Umeå 2030 Climate Neutrality Action Plan (Umeå Municipality, 2024); the Region Västerbotten Regional Innovation Strategy (Region Västerbotten, 2022); the Government of Sweden's 2024 industrialisation strategy; and EU-level documents including the Clean Industrial Deal and the 2024 Country Report for Sweden. Documentary triangulation grounds interview claims in observable facts (Patton, 2015; Yin, 2018).

3.5 Analytical Procedure: Joint Thematic Coding

The analytical procedure adopted in this study codes all materials together rather than separately. Treating the municipal interviews and the firm interviews as two parallel analytical halves a structure that an earlier exploratory phase of the work tested produced descriptions of institutional capacity and firm-level practice that ran

alongside each other without ever meeting, which is precisely the gap that a study of co-evolution cannot afford. The joint-coding procedure described below is closer to the recommendation of Braun and Clarke (2006, 2022) and consistent with how authors in the contemporary green path-development literature build findings sections (Trippl et al., 2020; Frangenheim, 2023; Jolly et al., 2025): themes are derived from joint reading of the whole corpus, and each theme is then evidenced by drawing on whichever data source most directly speaks to it.

The procedure followed the six standard phases familiarisation, initial coding, theme generation, theme review, theme definition, and writing up with the addition of a joint-coding step in which the two municipal transcripts, the five firm transcripts and the documentary corpus were read alongside one another and coded into the same set of analytical categories. Each individual transcript was first coded in detail using a per-interview coding table that records, for every coded segment, the surface-level code, the analytical category, the theoretical link, the unit of analysis, the role of the segment in the argument (driver, barrier, condition), and the direction of influence implied. These seven per-interview coding tables one each for the two municipal interviews and the five firm interviews are retained as the audit trail behind the consolidated matrix in Table 3.2 and behind the thematic argument in Chapter 5. Five themes emerged as recurring across the corpus and are reported in Chapter 5: (1) institutional supply and the dual face of municipal action; (2) the multi-scalar geography of co-evolution; (3) asymmetric co-evolution (who adapts to whom); (4) cultural toolkits as the interpretive hinge of co-evolution; and (5) procurement and finance as cross-cutting friction surfaces. The themes were stable across two rounds of coding.

Coding was carried out twice: a first inductive pass close to the data, and a second deductive pass mapping each code to the framework in Chapter 2. This combined

inductive-deductive logic is consistent with the abductive approach recommended for case-study work in the EEG and transitions traditions (Sotarauta, 2018; Coenen et al., 2017).

Table 3.2 sets out the consolidated coding matrix. Rows are the five themes that emerged from joint reading of the eight transcripts; columns are the eight data sources, the last of which is the anchor-firm interview (Firm F) defined in section 2.7. Each cell records the principal codes that the per-interview coding tables produced for that combination of theme and interview, abbreviated for compactness. Empty cells indicate themes that did not generate substantial codes in that particular interview. The anchor-firm column completes the matrix and shows, for each theme, how the large incumbent manufacturer's account compares with the municipal and MSME accounts.

Theme	Mun 1 (Bus. Dir.)	Mun 2 (Bus. Dev.)	Firm A (biological agri-tech)	Firm B (listed software)	Firm C (owner-led mfg.)	Firm D (marine tech)	Firm E (process tech)	Anchor (Firm F)
T1 - Institutional supply / dual face of municipal action	connector; intermediary; first contact; selective firm attraction; ecosystem of advisors; brokered governance	circular hub framing; Eco Industrial Park; industrial symbiosis; project-based pipelines; Industriråd dialogue	weak municipal support; no expected future support; municipality absent from firm experience	no proactive outreach; not engaged with municipality; municipality not central to listed-firm sustainability	politicians "talk but do opposite"; municipality builds new instead of restoring; local political mistrust	municipality "zero" role for exporters; no support function for export-oriented firms	operates outside municipal perimeter; no municipal planning impact on firm activity	"super good relationship"; municipality "listens to our needs"; eager to retain big industry; feels included and supported
T2 - Multi-scalar geography of co-evolution	EU/national framing of green transition; place-based attraction within national system	EU funding routes; Climate-Neutral Cities Mission; national programmes; regional partnerships	national regulator (Naturvårdsverket) is binding; cross-national EU regulatory mismatch	EU CSRD; Swedish stock-market listing rules; customer-country reporting; pan-European customer base	local-only framing through critique; national grants used (Klimatklivet, NV); RIS-distant	100+ export markets; EU grants; EKN; national export-finance gap; multi-scalar advocacy	40+ countries; Vinnova; EU grants; Business Sweden Catalyst; incubator-supported	Volvo Group strategy + global customers + EU regulation drive transition; "all of the above"; cabs/CKD steel for worldwide market; local plant, global system

Theme	Mun 1 (Bus. Dir.)	Mun 2 (Bus. Dev.)	Firm A (biological agri-tech)	Firm B (listed software)	Firm C (owner-led mfg.)	Firm D (marine tech)	Firm E (process tech)	Anchor (Firm F)
T3 - Asymmetric co-evolution (who adapts to whom)	firm must initiate ("say hello, I need help"); supply available but reactive; selective brokerage	only one MSME currently in Eco Park; municipality builds capacity faster than uptake	small firm powerlessness; "rely on big company"; cost equal but burden unequal; useless to fight rules	compliance with directives; rule-taker register; product-side influence on customers	self-financed transformation; owner ethics drive action; refuses dependence on institutions	15-20 year procurement and export-finance advocacy; "Don Quixote problem"; reverses direction	sustainability-embedded from founding; sectoral conservatism slows institutional uptake	municipality more responsive to large employer; receives attention smaller firms do not; but anchor constrained by group + 100,000-employee organisational stickiness; slow internal lead times
T4 - Cultural toolkit / interpretive register	"living circular hub" imaginary; mediation register; municipality as	circularity and industrial symbiosis as identity; project-based	customer-led, substance-oriented register; ecological	governance-and-reporting register; materiality; audit; product-	owner-driven ethical register; "do not just talk"; bicycles for employees; tree planting	transformative-enterprise register; sustainability-embedded; market	quality-led, networked-legitimacy register; greenwashing critique	premium-brand legitimacy register; "driving progress";

Theme	Mun 1 (Bus. Dir.)	Mun 2 (Bus. Dev.)	Firm A (biological agri-tech)	Firm B (listed software)	Firm C (owner-led mfg.)	Firm D (marine tech)	Firm E (process tech)	Anchor (Firm F)
	connector identity	municipal self-understanding	certification as identity	feature sustainability		education advocacy	("smoke and mirrors")	sustainability as brand trust; feels included but not direct Eco Park partner; no strong opinion on circular-hub framing
T5 - Procurement and finance frictions	acknowledges dense advisor ecosystem; mismatch with firm experience noted as comparison point	project funding pipelines (Climate City Contract); EU funding access through municipality	regulatory approval burden; complex rules disproportionate to micro-firm capacity; debt avoidance	CSRD compliance cost; reporting infrastructure investment as friction	Klimatklivet (EVs) easy; solar funding complex; first solar application rejected	procurement "pushes lowest price not sustainable"; EKN-bank disconnect; SEK 200M turnover threshold	procurement "paragraph on paper"; local SMEs lose to large Stockholm firms; price-pushing	binding conditions are infrastructure, reliable green electricity, and competence supply; needs sustainable north–south logistics; "we can't do it on our own" shared regional capacity

Table 3.2. Consolidated thematic coding matrix across all eight interviews, including the anchor-firm case (Firm F). Cells condense the codes recorded in the per-interview coding tables. Source: author.

3.6 Ethics, Anonymisation and Reflexivity

All participants were given an information sheet and gave consent before the interview began. The thesis followed Vetenskapsrådet (2017) guidelines. The municipal officials are identifiable by role, since the positions are public and unique within Umeå Municipality; they consented on that basis. The translator-mediated Firm A interview raises questions about voice, accuracy of translation, and family relationship; the coding stays close to the substantive content rather than over-interpreting specific phrasing. On anonymisation: all MSME firms and individual MSME interviewees are anonymised. The five completed MSME interviews are referred to as Firm A (a micro-scale green-niche firm in biological agri-tech), Firm B (a listed multi-unit software group headquartered in Umeå), Firm C (a small owner-led manufacturing firm of approximately twelve employees), Firm D (an export-oriented marine technology firm) and Firm E (a micro-scale process-technology firm with global reach). On reflexivity: the researcher is based at Umeå University, itself part of the RIS being studied. Documentary triangulation and the explicit Chapter 5 attention to where firm experience contradicts the municipal account are the practices used to manage this proximity (Berger, 2015).

3.7 Quality Criteria and Limitations

Following Lincoln and Guba (1985) and Tracy (2010), the thesis is assessed against four criteria. Credibility is supported by documentary triangulation, joint coding across all data sources, and explicit attention to where firm experience contradicts the municipal account. Transferability is supported by the thick description of Umeå's context in Chapter 4. Dependability is supported by the explicit description of the analytical procedure. Confirmability is supported by the requirement that every theme carries transcript evidence.

Four limitations should be acknowledged honestly. The study rests on a small and carefully chosen set of eight interviews, and that smallness is a choice rather than a shortcoming: the aim throughout has been to understand each case in depth rather than to count how often a pattern recurs across many firms, which is exactly the trade-off an embedded single-case design is meant to make (Yin, 2018; Flyvbjerg, 2006). The supply-side and demand-side of the story have been kept in balance by treating each MSME conversation as a way into a particular part of the municipal account, and the anchor-firm conversation as a way into the supplier-tier and Industriråd channels that the smaller firms could not speak to. One real constraint on that last conversation is that it was held with the plant's communications director rather than with the logistics or purchasing staff who handle suppliers day to day, so the supplier-tier mechanism comes through only partially (see section 5.7). A further limitation is timing: the policy environment has kept moving throughout the research. And the study works from interviews and documents rather than from sustained ethnographic presence, which means the cultural-toolkit perspective is read analytically rather than lived alongside the firms.

4. Case: MSMEs and Green Transition in Umeå

This chapter situates the case empirically. It describes Umeå's economic structure, the composition of its MSME sector, its sustainability and climate initiatives, and the multi-level institutional environment in which manufacturing MSMEs operate. The chapter draws principally on two business-sector analyses prepared for the municipality and the county (Frisk & Fineman, 2025; Fineman & Vieweg, 2026), on official municipal and regional documents, and on the EU-level policy frameworks within which all of this sits.

4.1 Umeå

Umeå is the largest urban centre in northern Sweden and the administrative seat of Västerbotten County. Its population has grown steadily over the past two decades and the municipality has set a long-term target of reaching 200,000 residents by 2050 within an explicit sustainability framing (Umeå Municipality, 2024). The city is one of seven Swedish cities selected for the EU Mission for 100 Climate-Neutral and Smart Cities by 2030 (Swedish Energy Agency, 2022) and has signed and renewed the Climate City Contract 2030 (Viable Cities, 2025). Umeå University, founded in 1965, is one of the largest universities in Sweden and a central actor in the regional innovation system, alongside the Swedish University of Agricultural Sciences (SLU). The city is well-connected by road, rail, and harbour, and it sits within the broader Norrbotten-Västerbotten geography that the Swedish government has identified as a priority for the country's industrial transformation (Government of Sweden, 2024).

The cultural dimension matters here, because it is not background but a substantive part of how local actors understand themselves and how the municipality projects the city's future. The theoretical framework in Chapter 2 takes the cultural-toolkit

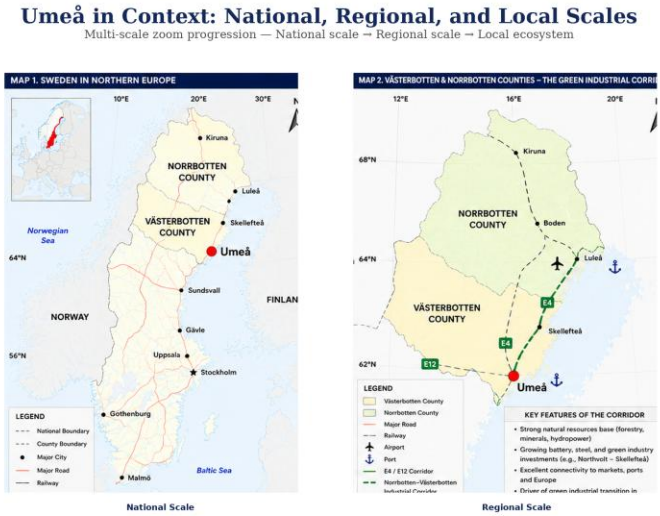
perspective seriously (Benner, 2026; Swidler, 1986), and the empirical purchase of that perspective depends on Umeå having a culturally legible identity. Three elements stand out. First, Umeå is part of Sápmi, the cross-border region historically inhabited by the Sámi, the only indigenous people recognised within the European Union. The Sámi name for the city is Ubmeje, and the Sámi presence is visible in the city's institutional life through Sámi research at Umeå University, the annual Sami Week (Ubmejen Biejvieh), and dedicated Sámi exhibitions at the city's museums. Second, Umeå was designated European Capital of Culture in 2014, the northernmost city in Europe ever to hold the title, with a programme that explicitly drew on the eight-season Sámi calendar as its organising structure. The 2014 designation built durable cultural infrastructure that continues to shape the city's self-presentation as a place where culture, university research, and sustainability ambitions are co-produced. Third, the local identity has long been organised around forestry and the river Umeå is known as the "City of Birches" after the avenues of birch trees planted following the 1888 fire and these ecological and material references continue to inform the cultural register in which the municipality now articulates its climate ambitions. The "living circular hub" framing analysed in Chapter 5 is best understood as the contemporary expression of this longer cultural repertoire rather than as a rhetorical innovation imposed from above.

This cultural legibility has a contested edge that should be acknowledged, even though it sits at the boundary of this study's scope. The same green industrial transformation that draws investment into northern Sweden also generates land-use conflict: the mines, wind farms, and transmission infrastructure the transition requires are frequently sited on lands used for Sámi reindeer husbandry, and the resulting tensions over land rights and consultation have become a recurring feature of the regional debate (Nordregio, 2024; Österlin & Raitio, 2020). Umeå itself is less

directly affected than the mining districts further north, but the wider Norrbotten-Västerbotten corridor within which the city sits cannot be described as a neutral green-growth space. Recognising this matters for the distributional reading the thesis adopts from Jolly et al. (2025): the question of who benefits from the green path, and on whose terms, extends beyond the firm tier to the Indigenous communities whose lands the transition reshapes. A full account of that dimension would require fieldwork beyond the present design, and it is flagged here and revisited in the conclusions as a boundary of the study rather than a claim it can settle.

Figure 4.1 situates the case geographically through a three-panel zoom progression that moves from national to regional to local scale. Map 1 (top left) places Umeå within Sweden and northern Europe, showing the city's position relative to Stockholm, Gothenburg, and the wider Nordic and Baltic context. Map 2 (top right) zooms in to Västerbotten and Norrbotten counties and renders the Norrbotten-Västerbotten industrial corridor that runs northwards from Umeå through Skellefteå to Luleå and Boden the corridor that hosts the larger flagship projects discussed in section 1.1 and that connects the case to ports, airports, and the E4/E12 transport network. Map 3 (below, shown at full width) zooms further in to Umeå Municipality itself and is the operational map for the empirical chapter, locating the verified positions of the institutional and industrial assets: Umeå University and SLU, Uminova Innovation, the Umeå Biotech Incubator, the major industrial manufacturing sites at JOST Umeå/Ålö, Komatsu Forest AB, Volvo Trucks Umeå, and Cytiva Umeå, the Umeå Eco Industrial Park at Dåva, Umeå Airport, and the Port of Umeå at Holmsund. Read together, the three panels show that the institutional architecture of Umeå is not a single point but a layered geography embedded in a strategic national and regional position, with the university cluster, the anchor

manufacturers, and the planned circular niche space at the Eco Industrial Park occupying distinct sites within the municipality.



Map 3. Key Industrial, Innovation and Institutional Locations in Umea

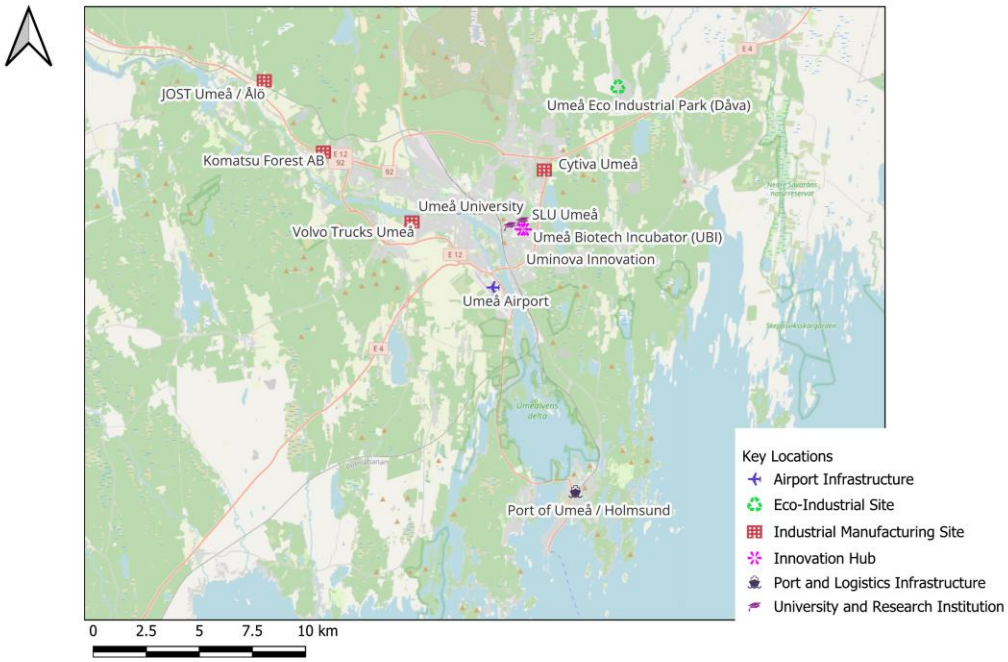


Figure 4.1. Umeå in context: from Sweden to local industrial, innovation and MSME ecosystem. Top: Map 1 (Sweden in northern Europe, national scale) and Map 2 (Västerbotten and Norrbotten counties and the green industrial corridor, regional scale). Bottom: Map 3 (Key industrial, innovation and institutional locations in Umeå, local scale, shown at full width for legibility). Sources: author's composition based on Lantmäteriet (2024), SCB (2023), Umeå Municipality (2024), Uminova Innovation (2024), and OpenStreetMap contributors (2024); locations on Map 3 verified against official addresses.

4.2 The Umeå Business Sector

Umeå's business sector has performed strongly in the years preceding this study. Between 2018 and 2023 the city's value added grew by 45%, against a Swedish national average of 37%; employment grew by 11% against 8% nationally; the number of firms grew by 16%, somewhat slower than the national rate of 24% (Frisk & Fineman, 2025). The local economy now counts roughly 6,667 firms and 36,400 employees in 2023.

The size-class composition of local employment is the analytically important fact for this thesis. Table 4.1 reports the size-class distribution as published by Dun & Bradstreet Sverige in their 2025 Umeå Näringslivsanalys, expressed both as a share of total local employment and as approximate absolute numbers. The percentages are the figures reported directly in the source (Frisk & Fineman, 2025, slide 40); the absolute numbers are the author's calculation, derived by multiplying each share by the 2023 total employment of 36,400.

Size class	**EC 2003/361/EC band**	**Share of Umeå employment (%)**	**Absolute employees (approx.)**	**Sweden share (%) for comparison**
0-9 employees	Micro	21%	~7,640	25%
10-50 employees	Small (10-49)	33%	~12,010	32%
51-100 employees	Medium (lower band)	14%	~5,100	13%
101-250 employees	Medium (upper band)	16%	~5,820	12%
MSME total (0-250)	**Micro + Small + Medium**	**84%**	**~30,580**	**82%**
251-500 employees	Large (lower)	7%	~2,550	6%

Size class	EC 2003/361/EC band	Share of Umeå employment (%)	Absolute employees (approx.)	Sweden share (%) for comparison
501+ employees	Large (upper)	10%	~3,640	12%
Large total (250+)	**Large**	**17%**	**~6,190**	**18%**
Total local employment (2023)	**All firms**	**100%***	**36,400**	**100%**

*Table 4.1. Distribution of Umeå employment by firm size-class, 2023. Percentages from Frisk and Fineman (2025, slide 40); absolute numbers calculated from the 2023 total of 36,400 employees. *Rows sum to 101% in the source due to rounding.*

Around 84% of local employment sits within the MSME range, which is a substantial structural fact: the city's economy is genuinely MSME-dominated and not just nominally so. Manufacturing has not collapsed. In 2023 the manufacturing sector still accounted for 35% of local employment, only marginally smaller than business-support services. The city hosts a small number of anchor firms in manufacturing in the sense of the term developed in section 2.7 operating in vehicle production, forestry machinery, life-science equipment, and process technology. These firms are referred to anonymously throughout this thesis in line with the anonymisation protocol described in section 3.6. Around them sits a substantial tier of smaller suppliers and processors. The most recent county-wide analysis confirms that even after the Northvolt bankruptcy, Västerbotten continues to expand: county employment growth between 2019 and 2024 remains close to 10% even when Northvolt-related firms are excluded (Fineman & Vieweg, 2026).

4.3 Green Transition Objectives and Policy Actors

Umeå's climate ambition is formally documented across several layered instruments. The municipality's own Climate Roadmap 2030 (UBC Sustainable

Cities Commission, 2023) and the Umeå 2030 Climate Neutrality Action Plan (Umeå Municipality, 2024) set out the local commitments. The Climate City Contract 2030, renewed annually under the Viable Cities programme, commits the municipality publicly to specific intermediate targets (Viable Cities, 2025). The Region Västerbotten Regional Innovation Strategy 2022-2030 (Region Västerbotten, 2022) sets out the broader county-level direction, while the Norrbotten-Västerbotten Industrialisation Strategy launched in June 2024 (Government of Sweden, 2024) frames the wider northern Swedish industrial transformation within which Umeå sits. EU-level instruments the Green Deal of 2019, the Net-Zero Industry Act and Critical Raw Materials Act of 2024, and the Clean Industrial Deal of February 2025 (European Commission, 2020, 2025a) provide the supranational landscape pressure that the framework labels.

The institutional architecture around manufacturing MSMEs is dense. At the municipal level, two functions matter most: the Head of Economic Development and the Business Developer responsible for industrial development. The municipality also runs a structured dialogue channel with major-firm CEOs called Industriråd. Beyond the municipality, the regional innovation system includes Umeå University and SLU; three incubators (Umeå Biotech Incubator, Uminova Innovation, and Expression); a public business-support agency (Almi); the regional government (Region Västerbotten); chambers of commerce and trade bodies (Företagarna, Svenskt Näringsliv); a green-technology cluster network (North Sweden Cleantech); and investment vehicles (Partner Invest). At the national level, the most important regulatory and finance actors for the firms in this study are Naturvårdsverket (Swedish Environmental Protection Agency) and the bodies responsible for SME export finance, including EKN and the Swedish Export Credit Corporation, which feature centrally in the Firm D interview. National innovation funding through

Vinnova, and internationalisation support through Business Sweden's Catalyst programme, feature in the Firm E interview. At the EU level, the Just Transition Fund, the Climate-Neutral Cities Mission, EU procurement directives, and EU innovation grants all define the formal opportunity space for the firms.

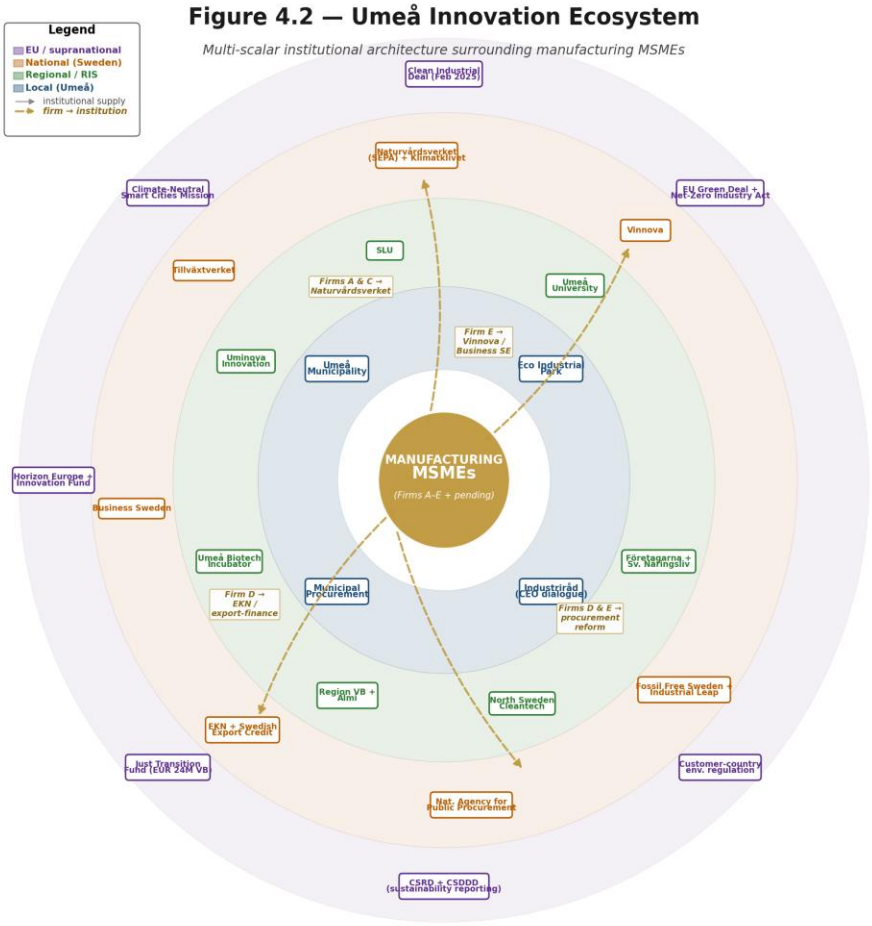
Two empirical features of this institutional landscape deserve particular emphasis because they recur in the findings. First, the Eco Industrial Park: a 550-hectare greenfield development with around 320 MW of electrical capacity, designed as a circular hub around industrial symbiosis. Current actors in the park include Umeå Energi, Umeå Deponi, INAB, Ragn-Sells, PreZero, Svevia, and Arevo; future establishments under discussion include Liquid Wind, Norrslam, Greenbirch, and Syntrophix. Second, the unusual combination of a strong supply-side institutional architecture and a not-yet-tested demand-side accessibility to MSMEs: the system is dense, but whether it is experienced as accessible is exactly what the firm-side interviews are designed to test. The next chapter takes this question up empirically.

Figure 4.2 visualises this institutional architecture as a concentric ecosystem around the manufacturing MSME tier. The diagram organises the actors named above into four scale-bands: local (Umeå Municipality, the Eco Industrial Park, the Industriråd, municipal procurement and planning), regional / RIS (the university and SLU, the incubators, Region Västerbotten with Almi and Partner Invest, North Sweden Cleantech, the trade bodies), national (Vinnova, Naturvårdsverket, Tillväxtverket, Business Sweden, EKN and Swedish Export Credit, the National Agency for Public Procurement, Fossil Free Sweden), and EU / supranational (the Green Deal, the Clean Industrial Deal, the Climate-Neutral Cities Mission, Horizon Europe, the Just Transition Fund, the Corporate Sustainability Reporting Directive [CSRD] and Corporate Sustainability Due Diligence Directive [CSDDD], and customer-country environmental regulation). The December 2025 Omnibus simplification agreement,

approved by the European Parliament on 16 December 2025, narrowed the scope of both CSRD and CSDDD substantially: the CSRD now applies only to companies with more than 1,000 employees and more than EUR 450 million in net turnover, and the CSDDD only to those with more than 5,000 employees and more than EUR 1.5 billion in net turnover, with CSDDD application deferred to July 2029 (Council of the European Union, 2025). In practical terms this means that no MSME falls directly within either instrument at the post-Omnibus thresholds; pressure on Umeå's manufacturing MSMEs comes instead through value-chain reporting requests from larger in-scope customers a point developed empirically in Chapter 5.

Figure 4.2 — Umeå Innovation Ecosystem

Multi-scalar institutional architecture surrounding manufacturing MSMEs



Source: author, based on Interviews 1–7 and Region Västerbotten (2022), Government of Sweden (2024), European Commission (2025a), Viable Cities (2025).

Figure 4.2. The Umeå innovation ecosystem for manufacturing MSMEs. Multi-scalar institutional architecture (local · regional · national · EU) surrounding the manufacturing MSME tier. Inward arrows show institutional supply; gold dashed arrows show co-evolutionary feedback from firms back to institutions. Source: author, based on Interviews 1-7 and documentary sources.

5. Findings and Discussion

This chapter reports and interprets the empirical findings of the joint thematic coding described in section 3.5. Following the analytical logic set out there, the findings and their discussion are presented together rather than in two separate chapters: each theme is first evidenced from across the corpus and then read against the theoretical framework, so that the empirical pattern and its co-evolutionary interpretation sit side by side. The five themes set out below emerge from reading all eight interviews two municipal, five MSME, and one anchor-firm together with the documentary corpus, not from analysing the institutional, firm-side and anchor-firm material in separate sections. Each theme is presented with direct quotations from across the data, organised so that the municipal account, the MSME accounts and the anchor-firm account can be heard alongside one another. The chapter then develops the cross-cutting interpretation, with section 5.7 tracing co-evolution as a process through five case sequences.

5.1 Reading All Data Sources Together

Joint coding of two municipal interviews, five MSME interviews, one anchor-firm interview, and the documentary corpus produced five recurring themes. The first is the dual face of municipal action the way Umeå Municipality presents itself simultaneously as a strategic connector and as an operational builder. The second is the multi-scalar geography of co-evolution the consistent finding that the institutions that matter most to firms' green transformation are rarely at the municipal scale. The third is the asymmetric direction of co-evolution the fact that, across the corpus, firms more often adapt to institutions than the other way round, and the conditions under which that asymmetry is reversed or, in the anchor-firm case, partially inverted. The fourth is the cultural hinge the way shared interpretive registers (or their absence) determine whether the institutional channels described by the

municipality are recognised by the firms as relevant to their work. The fifth is the cross-cutting friction surface of procurement and finance and, for the anchor firm, the binding regional conditions of infrastructure, electricity and competence where the process of co-evolution either advances or stalls. These five themes are presented in turn below; section 5.7 then traces how co-evolution unfolds over time across five case sequences, drawing the themes together into a process account.

5.2 Institutional Supply and the Dual Face of Municipal Action

The two municipal interviews, read alongside the Climate City Contract and the Eco Industrial Park documentation, show that the municipality acts on the local economy through two complementary registers. The first is strategic-relational. The Business Director frames himself in plain terms as an intermediary:

> *"It's kind of, I would say, to be a connector between the companies and the municipality. And of course, between the companies, not only the municipality, but the Umeå in a whole ... we're kind of what's called intermediary between companies and the rest of the Umeå."*
>
> *- Business Director, Interview 1*

This is closer to what the EEG and RIS literatures have begun to call relational governance: institutional action that operates through mediation, dialogue and connection rather than through rule-making (Gong & Hassink, 2019; Benner, 2022, 2026). The municipality also exercises selective firm attraction. The Business Director's account of who they look for is explicit:

> *"Now we're looking for companies that are contributing in the waste area, I mean, for plastic circularity and so on. So first of all, we have to have kind of a need for some companies or something else in the chain of manufacturing."*
>
> *- Business Director, Interview 1*

The second register is operational-material. The Business Developer describes municipal action through projects, sites and infrastructures. The Eco Industrial Park is the centrepiece, and she frames its purpose in cultural-imaginary terms as much as in physical ones:

> *"We have put up goals for development in the park ... we want to be this circular hub working together with universities, business and municipality together, and we work within principles of circularity, industrial symbiosis that means that one rest stream can be a benefit for another business."*
>
> *- Business Developer, Interview 2*

The two perspectives are not in conflict; they are layered. Together they describe what the municipality actually does: it brokers entry and it builds the material conditions that make that entry meaningful. This is the dual face of municipal action, and it runs through the municipal sequence traced in section 5.7 (Trace 4).

But the supply-side account on its own has a limit, and both interviewees explicitly flag it. The Business Director, asked whether the support landscape is accessible to MSMEs, draws the careful distinction:

> *"We have an ecosystem of advisors that can be ... if your company asks, well, we need actually need help to make a transition. There are abilities, but you have to actually ... say, hello, I need some help here."*
>
> *- Business Director, Interview 1*

The Business Developer reaches the same point through a different route. When asked how many MSMEs are currently being directly cooperated with in the Eco Industrial Park, her answer is one (Syntrophix). The supply-side architecture is real and dense; whether it actually reaches the manufacturing MSME tier is exactly the question the firm-side data are designed to test.

The anchor firm (Firm F) supplies the sharpest contrast in the corpus on this theme. Where the MSMEs experience municipal action as distant or absent, the anchor firm describes a close and responsive relationship. Asked how the firm relates to the municipality, the communications director is unambiguous:

> *"I wish we have a super good relationship … the municipality, they are really listening into our needs. We feel that they are quite eager to continue to have a big industry within the region. And we feel that we have the support that we need in the questions that are important for us."*
>
> *- Communications Director, Firm F, Interview 8*

This is the same municipality that Firm D described as having a "zero" role and that Firm E described as operating "outside the perimeters" of its work. The divergence is not a contradiction in the data; it is the finding. The municipal architecture is experienced as close and enabling by the large anchor employer and as distant or irrelevant by most of the smaller firms. This asymmetry a single institutional apparatus that reaches the anchor firm but not the MSME tier around it is the empirical hinge into the second theme, and it recurs through the rest of the chapter.

5.3 The Multi-Scalar Geography of Co-evolution

The most consistent finding across the five firm-side interviews is that the institutions that matter most to the firms' green industrial transformation sit somewhere other than the municipal scale. The point is made in five distinct ways. Firm D the green marine technology firm that exports to over one hundred countries is the most direct:

"From our perspective, the Umeå municipality has nothing to do … they don't have anything to do with exporting companies, so zero."

- CEO, Firm D, Interview 6

Firm B, the listed software group, draws an only slightly softer line. Asked whether the municipality has supported their green transformation, the investor-relations lead's answer is:

"Nothing for us … No support. And to be honest, we haven't asked for it either, but there has been no proactive outreach to us as far as I'm concerned."

- Investor-relations lead, Firm B, Interview 4

Firm B's account is sectorally specific. The firm is physically embedded in Umeå as employer (around 80 staff at headquarters) but it is not embedded as a customer-served firm most of its 26,000 customers sit elsewhere in Europe. The binding regulatory pressure on Firm B is the EU directives on sustainability reporting, the Swedish stock-market listing requirements, and indirectly the sustainability goals of customers that the firm's software helps them meet. The investor-relations lead is unambiguous about where the pressure originated:

"To be honest, I would say what … actually was the pressure we are a listed company on the Swedish stock market and there were the regulations that we needed to do this … it was also a pressure on listed companies to follow these rules."

- Investor-relations lead, Firm B, Interview 4

Firm A's scale story is national rather than EU. The firm's green-industrial-transformation move substituting a biodegradable carrier for the previous non-biodegradable one in its biological plant-protection product was driven by customer demand. The institutional friction that mattered, by contrast, was national:

"The Swedish agency for environment they are the company that stops a lot of the changes ... you must have approval to use these pesticides, these nematodes, and they are not approving ... she feels they are the biggest challenge."

- Owner-founder (translator-mediated), Firm A, Interview 3

Firm E, the process-technology micro-firm, frames the multi-scalar geography of its work even more explicitly. The interviewee draws the line between the municipality not relevant and the wider Regional Innovation System:

"We operate outside the perimeters of the local municipality and their planning and so forth. So their planning does not affect us at all."

- Business Manager, Firm E, Interview 7

But the same firm describes its survival path through a sequence of regional-and-national supports stretched over 15 years:

"For the first 15 years of this company's life, we survived on grants, we survived on funding, as well as money from our investors ... incubator support, startup support through Vinnova as an example, EU grants and so forth."

- Business Manager, Firm E, Interview 7

Firm C is the only one of the five whose co-evolutionary frame is partly local and even there, the frame is one of critique rather than active engagement. The owner-CEO comments on the consistency of local political commitments and describes his own action as proceeding regardless of them:

> *"The political … they talk about it, but still they build new houses. And I don't think that they should do new houses. They should restore and don't tear down concrete houses … they talk, but they do the opposite way."*
>
> *- Owner-CEO, Firm C, Interview 5*

Read jointly, the five firm-side accounts converge on a common pattern: the scale at which firm-side co-evolution actually happens is mostly not local. Firm A's binding co-evolutionary relationship is with Naturvårdsverket at the national scale; Firm B's is with EU sustainability-reporting directives and with its customers' national-level reporting obligations; Firm D's is with Swedish national export-finance institutions and with EU procurement and innovation programmes; Firm E's is with Vinnova, with EU programmes, and with Business Sweden. Only Firm C has a co-evolutionary frame primarily articulated against local political conditions, and even there the frame is one of critique. This pattern sits in tension with the local and regional bias of much evolutionary economic geography, which has conventionally treated the region as the privileged scale at which path dependence accumulates and at which new paths are created (Boschma & Frenken, 2006; Martin & Sunley, 2006). The Umeå corpus suggests that for export-oriented manufacturing MSMEs the path-shaping institutions are increasingly extra-regional national, EU and customer-driven so that the regional scale, while it is where the firms are embedded as employers and producers, is not where their green path is principally determined.

The anchor firm extends this pattern rather than breaking it. Although it is the most locally embedded actor in the corpus nearly a century of regional history and around 1,800 local employees its green transformation is driven from several scales at once. Asked what pushed the firm's changes forward, the communications director answers in explicitly multi-scalar terms:

> *"I would say it's all of the above. Of course, there are regulations that we need to follow ... and then the customer demand is also one of the biggest drivers ... we're a premium brand and we need to be ahead and we need to show that we are driving the change."*
>
> *- Communications Director, Firm F, Interview 8*

The anchor firm's production sits in Umeå, but its sustainability direction is set by the group, by EU regulation, and by customers worldwide. The plant manufactures cabs and pressed steel that travel into a global product system; as the communications director puts it, the steel "travels north" to be pressed and the finished cab "travels south again" to chassis plants in Gothenburg and Belgium. The firm is locally embedded but not locally determined the same multi-scalar geography the MSMEs describe, observed from the position of a large incumbent. This is the central comparative finding of the theme: across the entire size range of the corpus, from a one-person micro-firm to an 1,800-employee anchor plant, the institutions that bind green industrial transformation are predominantly national, EU and customer-driven rather than municipal. It is a strong confirmation of the open, multi-scalar reading of regional innovation systems (Isaksen, Trippl & Mayer, 2022) and of Frangenheim's (2023) interpath-across-regions argument the proposition that a region's green path is not produced in isolation but is shaped by, and develops through, its connections to paths unfolding in other regions, so that what looks like a single local trajectory is in fact assembled across several regional and national settings at once.

5.4 Asymmetric Co-evolution: Who Adapts to Whom

If the second theme concerns where co-evolution happens, the third concerns its direction. The theoretical framework treats co-evolution, following Gong and Hassink (2019), as a directional relationship that can be one-way, two-way, or

asymmetric. The corpus shows that the more interesting empirical question is when and how the direction is reversed.

In four of the five cases, the dominant direction is institution-to-firm: firms adapt to regulatory, customer, or procurement signals more than the other way round. Firm A adapts to Naturvårdsverket's product-approval procedures; Firm B adapts to EU sustainability-reporting directives and the Swedish stock-market listing regime; Firm E adapts to the slow regulatory pace of pharmaceutical and nuclear adoption decisions; Firm C's self-financed investments in solar, electric vehicles and geothermal heating are not in any direct sense an adaptation to institutional pressure at all (the owner-CEO insists on this explicitly), but the direction is still firm-internal rather than firm-on-institution. Firm A's account of how a small firm tries and fails to shift the regulator captures the asymmetry vividly:

> *"Because we are a smaller company, we rely on the big company to make the effort to approve the new things for us. We can't afford to change the government or the agencies… The cost is the same, but it's a small cost for a big company, but it's a huge amount of money for us."*
>
> *- Owner-founder (translator-mediated), Firm A, Interview 3*

Firm D is the corpus's clearest counter-case. The firm is actively trying to reshape three institutional terrains: market evaluation criteria (advocating LCOE levelised cost of energy over CAPEX/OPEX in floating-solar markets), public procurement rules, and national export-finance architecture. The interviewee describes the procurement effort in concrete terms:

> *"If you look at the Umeå commune when they're buying, for instance, lamps and pencils … they are still pushing lowest price. They are not*

pushing sustainable purchases. That is the first starting point where they need to change that."

- CEO, Firm D, Interview 6

On the national side, the same firm describes a sustained, repeated effort over many years to alter the Swedish export-finance architecture, including direct dialogue with the responsible national minister:

"That is the reason we met here with [the minister] … he understands this problem and would want to make a change, but we'll see. I've been struggling with this since perhaps 15, 20 years trying to get this. This is my Don Quixote problem."

- CEO, Firm D, Interview 6

Firm D, in other words, is the corpus's strongest case of firm-led co-evolution. It is also importantly a sustainability-embedded firm rather than a sustainability-adapted one. The interviewee is emphatic that the firm did not transform toward sustainability; it was on the green path from its founding decades ago, and the rest of the market is only now catching up:

"Those are core fundamental values of the reason why we're pushing this as a solution … we've been pushing these two things, sustainability and environmental aspects, from the very beginning."

- CEO, Firm D, Interview 6

Firm E shows a softer version of the same pattern. The firm describes its sustainability claim as foundational and predating the market's arrival at it:

"Sustainability has always been at the core of what we do and the how the business has developed … we now [are] having a larger and larger impact on the end environment with the end customer."

- Business Manager, Firm E, Interview 7

The anchor firm complicates the asymmetry in two instructive ways. First, it shows that organisational scale cuts both ways. A large firm has resources the MSMEs lack, but it also carries an organisational stickiness that the small firms do not. The communications director draws the contrast directly:

> *"It's quicker if you are a small company, if you have 5 or 10 or 20 employees … you gather those that can take the decisions and you just decide and then you go ahead. The process for a company with 100,000 employees … takes longer time."*
>
> *- Communications Director, Firm F, Interview 8*

This is path dependence operating inside a large firm rather than across a region: the very scale that makes the anchor firm institutionally visible also slows its internal adaptation. Second, and most importantly for the direction question, the anchor firm receives a quality of municipal attention that the MSMEs do not. The municipality is described as "eager to continue to have a big industry within the region" and as listening closely to the firm's needs. The asymmetry of co-evolution is therefore not only about firms adapting to institutions; it is also about institutions adapting more readily to some firms than to others. The municipality is more responsive to a single large anchor employer of 1,800 people than to the many small firms that, in aggregate, employ far more. This is the distributional edge that Jolly et al. (2025) insist green path-development research must examine: whose voice is institutionalised, and on what terms. In the Umeå corpus, the anchor firm's voice is the one the municipality most readily hears.

5.5 Cultural Toolkits as the Interpretive Hinge of Co-evolution

The fourth theme is cultural. The framework treats Benner's (2026) toolkit perspective as the layer that determines whether the formal channels of co-evolution

are recognised as relevant by the actors who could in principle use them. The Umeå corpus shows the toolkit doing exactly this analytical work.

The municipal cultural register is the "living circular hub" framing the Business Developer articulates. It is not rhetoric; it is the interpretive vocabulary within which she and her colleague describe their own work. But the five firm-side interviews show that this register is only partially recognised by the firms it is designed to engage. Firm D operates in a transformative-enterprise register that draws on lifecycle cost, market education and procurement reform; this register is closer to the municipal one in spirit but reads its own market work as outrunning the municipal frame. Firm E speaks through a quality-led, networked-legitimacy register openly sceptical of greenwashing:

> *"A lot of the green transition is also smoke and mirrors. There's a lot of façade, there's a lot of front, there's a lot of ticking boxes. We're actually about making an actual change."*
>
> *- Business Manager, Firm E, Interview 7*

Firm C's register is owner-driven and ethical. The investments solar, electric vehicles, geothermal, green aluminium, tree-planting, employee bicycles are framed as personal moral commitments rather than as participation in a municipal project. The follow-up interview adds a striking detail of how this register plays out in everyday practice:

> *"In 2020 we actually purchased bicycles for the employees, because we don't want the employees to just drive back and forth … we invested in or purchased bicycles for all the employees so they could take the bike to work instead of driving all summer."*
>
> *- Operations lead, Firm C, Interview 5 (follow-up)*

Firm A's register is customer-led and substance-oriented; the substantive content of sustainability for the firm is the specific material composition of the carrier in which the biological agent is held, and the relevant external actors are customers asking for an ecological product rather than the municipality projecting a circular city. Firm B's register is governance-and-reporting; sustainability for the group is described through the vocabulary of materiality, audit, customer-sustainability goals and product-feature innovation.

The anchor firm adds a sixth register: premium-brand legitimacy. Sustainability, for the anchor firm, is bound up with being seen to lead with the brand promise of "driving progress" and with the trustworthiness a premium manufacturer must project. Yet when the municipal "living circular hub" imaginary is put to the communications director directly, the response is telling. The firm feels included in municipal forums, but it does not adopt the municipal frame as its own:

> *"We feel that we are included. But the eco industrial park in Dåva, it's not something that we are … we are not the partner in that development … it's not something that we have opinions on their framing."*
>
> *- Communications Director, Firm F, Interview 8*

This is the most analytically important cultural finding in the corpus. The anchor firm has the closest operational relationship with the municipality of any firm interviewed and yet it, too, does not reach for the circular-hub imaginary. Inclusion in municipal forums and a strong working relationship do not produce shared cultural framing. The convergence finding across all six registers five MSME registers plus the anchor firm's premium-brand register is that none of them naturally adopts the "living circular hub" vocabulary. This is not a failure of municipal communication. It is a structural feature of how firm-level sustainability-meaning is made when firms answer to customers, regulators, owners and group strategies

distributed across multiple scales. The cultural register is the hinge between institutional supply and firm-side recognition, and in Umeå the hinge is only partially engaged even for the firm that sits closest to the municipal apparatus.

5.6 Procurement and Finance as Cross-Cutting Friction Surfaces

The fifth theme is the most concrete. Across the corpus, two cross-cutting frictions appear as the practical points at which the process of co-evolution either advances or stalls: public procurement and SME export finance.

On procurement, Firm D and Firm E converge from different sides of the same problem. Firm D, as a supplier of sustainable solutions, describes the procurement default as the structural barrier:

> *"The basic fundamental problem is that the general community doesn't differentiate between price and cost … a sustainable product can be priced higher. That does not mean that the cost is higher … if you have three times the lifespan, the total cost of ownership is much less than the product that has a lower price."*
>
> *- CEO, Firm D, Interview 6*

Firm E names the same problem from the local-SME side:

> *"It's very seldom that a local small SME will win the procurement contract, even though it is the most advantageous. There will be some large national firm from Stockholm that will win it because they can push down prices … there is a sustainability factor in all the procurement contracts. However, from what I've heard and experienced, it appears to simply be a paragraph on paper."*
>
> *- Business Manager, Firm E, Interview 7*

Two interview accounts, two corners of the corpus, one institutional bottleneck. The National Agency for Public Procurement is identified by Firm D as the body that could shift this, and lifecycle-cost methodology is named as the technically available alternative.

On finance, Firm D's account of the Swedish export-finance gap is the corpus's most specific institutional barrier:

> *"EKN ... would happily lift the risks. But the idea was then to take that invoice and go to one of the commercial banks here ... that doesn't work here in Sweden ... there is more aligned and more available network for SMEs to find export financing [in other countries]."*
>
> *- CEO, Firm D, Interview 6*

> *"For larger companies having more than 200 million Swedish kroner turnover per year ... they can go to the organisation called SEK, Swedish Export Credit ... they can go there and find export finances for their export sales ... if you have a turnover less than 200 million per year, then that door is closed."*
>
> *- CEO, Firm D, Interview 6*

The SEK 200 million threshold draws a structural line through the MSME band. Two further frictions are visible across the corpus but did not crystallise as standalone themes. The first is application complexity: Firm C's description of Klimatklivet-style support for electric vehicles is positive, but its description of the equivalent solar-panel funding process is markedly more difficult:

> *"Now it's easy and it's called Naturvårdsverket [for the EV grants], and that applies very easy. But for the solar energy ... it was much more complicated. The first time we did the one solar energy [installation] we got no money because we did wrong in the early process."*

- Owner-CEO, Firm C, Interview 5

Firm A reports an analogous pattern: regulatory product-approval procedures designed for established chemical products do not fit biological products with a shorter market history, and the complexity costs are disproportionately heavy for a one-employee company. The second friction is competence the Business Developer's recurring point that finding the right people is the binding constraint on growth in northern Sweden.

The anchor firm brings the competence friction into sharp focus and adds two more binding conditions that the MSME cases touch only lightly: infrastructure and electricity. Where the smaller firms experience friction at the procurement and finance surfaces, the anchor firm's binding conditions are the physical and human prerequisites of remaining in Umeå at all. The communications director names them precisely:

> *"There are three things that are super important to us. Number one is infrastructure. If we can't deliver our products, then there's no reason for us to have a plant in Umeå … number two is the electricity demands … we can't afford even the smallest dips … And then number three is the competence."*
>
> *- Communications Director, Firm F, Interview 8*

These three conditions sustainable transport infrastructure, reliable green electricity for a highly automated plant, and a supply of specialised competence are exactly the conditions that municipal and regional institutional capacity is positioned to provide. They are also explicitly framed as a shared, co-evolutionary responsibility rather than something the firm can secure alone:

> *"We can't do it by our own, but we can solve it together ... within the region, within the group. If we just work together towards common goals, then ... this would be possible."*
>
> *- Communications Director, Firm F, Interview 8*

This is the clearest single statement of co-evolution in the corpus: a large firm naming the institutional conditions of its own continued green operation and casting them as a joint regional project. It also reframes the friction-surface theme. For MSMEs the friction surfaces are procurement and finance; for the anchor firm they are infrastructure, electricity and competence. In both cases the friction is where firm-level transformation meets the limits of what the firm can provide for itself, and where institutional capacity at the right scale either unblocks the process or lets it stall.

The procurement-and-finance friction surface, and the anchor firm's infrastructure-electricity-competence conditions, are where the Umeå case shows the process of co-evolution most concretely advancing or stalling. The interpretation of these findings across all five themes, and the policy implications that follow, are developed in the remainder of this chapter and drawn together in the conclusions.

5.7 The Process of Co-evolution: Five Case Sequences

The five themes describe what co-evolution looks like in aggregate; they do not yet show how it unfolds over time. This section traces the process directly. It takes up the question Frangenheim (2023) and Jolly et al. (2025) press, namely how co-evolution actually proceeds phase by phase, through which channels, and at what cadence, and reconstructs five sequences from the joint coding tables of section 3.5: three firm-side (Firms A, D and B), one institution-side (the municipality), and one anchor-firm (Firm F). The three firm-side traces were chosen because each instantiates a distinct pattern, stalled, firm-led reversal, and customer- and directive-

led; Firms C and E fall within the same patterns and are not traced separately, to avoid repetition.

Trace 1 - Firm A and Naturvårdsverket: stalled co-evolution at the national regulatory scale. The firm reformulates its biological plant-protection product onto a biodegradable carrier in response to customer demand, then presents it to a national approval regime built around chemical pesticides rather than biological agents. Lacking the legal capacity or the financial reserves to push the regulator into procedural adaptation, "it feels useless for a small company to fight the rules," the firm stalls at the institutional-engagement phase. The regulatory-approval channel is technically active but practically blocked: each side has acted, but neither has shifted the other.

Trace 2 - Firm D and the national export-finance architecture: long-run, firm-led co-evolution still in motion. Sustainability-embedded from founding and now exporting to more than one hundred countries, the firm runs into the mismatch between EKN's invoice-risk guarantees and the working capital that Swedish commercial banks decline to provide for smaller exporters, and into the SEK 200 million Swedish Export Credit threshold that draws a size-based line through the MSME band. Fifteen to twenty years of advocacy, including direct ministerial dialogue, have reached a minister who "understands this problem and would want to make a change, but we'll see." Direction is reversed, the firm acting on the institution, at national and EU scale and through advocacy rather than procurement, with institutional change anticipated but not yet realised; the multi-decade cadence illustrates Jolly et al.'s (2025) point that whose voice gets institutionalised depends on accumulated firm capacity over time.

Trace 3 - Firm B and the EU sustainability-reporting regime: customer- and directive-led co-evolution at the EU scale. The CSRD and CSDDD bind the firm

both directly and, through customers who must report on value-chain emissions, indirectly; the firm folds materiality assessment, reporting infrastructure and audit-readiness into customer-facing product features, and its pan-European customer base, not Umeå, drives the direction. By late 2025 the Omnibus simplification narrows the directives' scope (Council of the European Union, 2025), shifting the pressure on smaller firms toward indirect, value-chain reporting requests. Direction runs institution-to-firm and then firm-to-customer, at EU scale, in a governance-and-reporting register, illustrating Frangenheim's (2023) interpath-across-regions argument from the firm side.

Trace 4 - Umeå Municipality and the firm tier: institution-side co-evolution. The municipality moves from strategic framing (the "living circular hub" imaginary) through asset assembly (the Eco Industrial Park at Dåva) to governance routinisation (the annually renewed Climate City Contract and the Industriråd dialogue with major-firm CEOs), and on to selective firm attraction toward firms that fit the imaginary. The present phase is partial uptake: one MSME is directly engaged in the Park, while most of the interviewed manufacturers are not engaged with the municipal architecture they sit inside. Direction runs supply-to-demand at a durably local scale; the municipality has built more channels than the firms have entered, and its cultural register has stabilised into an imaginary that the firm side recognises only partially.

Trace 5 - the anchor firm (Firm F) and its multi-scalar environment: locally embedded but not locally determined. Made possible by the eighth interview, this trace combines deep local embeddedness with strongly non-local direction. A cab and pressed-steel plant with almost a century of regional history and around 1,800 local employees pursues sustainability on two linked tracks: greening its own highly automated production (a recent agreement for fossil-free, traceable electricity) and

greening its products (battery-electric alongside combustion vehicles, with a 2040 fossil-free target set against 2019 and 2030 milestones). It names regulation, customer demand and group strategy as jointly responsible, with premium-brand reputation an added driver. Its three binding conditions, infrastructure, reliable electricity and competence, are the conditions on which the plant's continued operation in Umeå depends, and the firm frames them as a shared regional project ("we can't do it by our own, but we can solve it together"). It describes a strong, responsive municipal relationship yet does not adopt the circular-hub framing and is "not the partner" in the Eco Industrial Park; it would welcome a denser northern supplier base, a "dream" of a north-to-south flow, but is "not there yet," and the detailed supplier-tier mechanism sits with logistics, purchasing and engineering rather than with the communications function interviewed (see section 3.4). Trace 5 thus shows co-evolution running simultaneously from local plant to global product system, predominantly group-and-market-to-firm with a strong firm-to-region feedback on the binding conditions, in a premium-brand register that coexists with but does not merge into the municipal imaginary.

Read together, the five sequences form a single co-evolutionary system rather than five separate stories. The municipal trace is structurally distant from Firm A's national regulatory bottleneck, selectively connected to Firm D's procurement-reform argument, and disconnected from Firm B's EU-and-customer pathway; its strongest interlink is with the anchor firm, through the Industriråd and the shared binding conditions, yet even there the supplier-tier channel that would extend the municipality's reach into the MSME tier is desired but not yet operative. Co-evolution between Umeå and its manufacturing tier is therefore multi-channel, asymmetric and partly absent: most present in the anchor-firm relationship, partially open in the procurement window, and structurally absent for the firms whose

transformations run through national and EU channels. What the sequences add to the thematic account is the temporal grain of the process: the phases through which it moves, the very different cadences at which it runs, from a single product-reformulation cycle to multi-decade advocacy, and the points at which it feeds back or stalls.

6. Conclusions

This chapter draws together the contributions of the thesis on the basis of the full eight-interview corpus two municipal interviews, five manufacturing-MSME interviews, and one anchor-firm interview analysed through joint thematic coding and interpreted in Chapter 5. It begins by answering the research questions directly, then states the empirical, theoretical and policy contributions, and closes with limitations and directions for future research.

6.1 Answers to the Research Questions

The thesis posed one main research question and three sub-questions. Each is answered here in turn, drawing the relevant evidence and interpretation together from Chapter 5.

Main research question

How does green industrial transformation among manufacturing MSMEs in Umeå co-evolve with local institutional capacity and planning systems? The answer the thesis reaches is that co-evolution does occur, but that it is multi-scalar, asymmetric, and culturally mediated, and that it runs only partly through the local scale. Across the whole size range of the corpus from a one-person micro-firm to an 1,800-employee anchor plant the institutions that most bind firms' green transformation are predominantly national (Naturvårdsverket, EKN, Swedish Export Credit), EU-level (the CSRD and CSDDD reporting regime, Horizon and procurement directives), and customer- or group-driven, rather than municipal. Local institutional capacity in Umeå is dense and genuinely transformation-oriented, but its co-evolutionary relationship with the manufacturing tier is uneven: strong and mutually acknowledged with the anchor firm, partially open with one or two larger MSMEs, and structurally thin with the smaller firms whose binding relationships run

elsewhere. Co-evolution, in short, is real but distributed across scales, and the local scale is one node in that distribution rather than its centre.

Sub-question 1

How do manufacturing MSMEs in Umeå including both incumbent firms and green start-ups engage in green industrial transformation through changes in production practices, technologies, or business models? The corpus shows that they engage through markedly different modes, and that the mode depends on the firm's founding orientation and market position more than on local conditions. Firm A engages through product reformulation (substituting a biodegradable carrier for its biological agent); Firm B through business-model and governance change (materiality assessment, sustainability reporting, and the folding of reporting capability into its product); Firm C through self-financed, owner-driven practice change (solar, electric vehicles, geothermal heating, employee bicycles); Firm D through sustainability-embedded product design carried from founding and propagated to export markets; and Firm E through sustainability-embedded process technology with global reach. The incumbent/green-start-up distinction drawn in Chapter 1 holds, but the more analytically useful distinction the data reveal is between sustainability-adapted firms (which change toward a green path under external pressure) and sustainability-embedded firms (which were founded on a green path and now find the market and the regulators slowly arriving at it).

Sub-question 2

How do local institutional conditions including planning systems, support structures, and policy frameworks influence MSMEs' ability to pursue green transformation? The finding here is, on the face of it, a low-influence finding but a low-influence finding is itself a substantive answer, and the thesis reads it as such. Local institutional conditions influence MSMEs' green transformation less by directly

shaping firm behaviour than by structuring which firms become visible to municipal action in the first place. The dense supply-side architecture the Eco Industrial Park, the incubators, North Sweden Cleantech, Almi, the climate roadmap, the Climate City Contract is real, but its direct influence on the five MSMEs is limited: most describe municipal action as distant ("outside the perimeters"), reactive (available only if the firm initiates contact), or irrelevant to their binding constraints. The stronger local influence is indirect and selective: the municipality is markedly more responsive to the large anchor employer than to the smaller firms, so local institutional conditions shape the distribution of attention and visibility across the firm tier even where they do not shape firm behaviour directly. The binding influences on firms' ability to transform regulatory approval, procurement rules, export finance, sustainability-reporting obligations run at national and EU scales. The clearest case of strong local institutional influence is the anchor firm, whose binding conditions (infrastructure, electricity, competence) are precisely the conditions municipal and regional capacity can provide, and which therefore experiences a local co-evolutionary relationship that the MSMEs do not.

Sub-question 3

How do interactions between MSMEs and municipal institutions contribute to continuity or change in local industrial development pathways? The interactions that exist contribute mostly to continuity of the current path rather than to its change. The municipality's selective firm-attraction (toward firms that fit the circular-hub imaginary), its Eco Industrial Park, and its Climate City Contract reproduce and extend an existing trajectory; the one channel most likely to drive pathway change the propagation of anchor-firm sustainability requirements into a denser regional supplier tier through the Industriråd is desired by the anchor firm but not yet operative. The anchor-firm interview confirms that this change-bearing channel

exists and is wanted (the firm's "dream" of a north-to-south flow with more northern suppliers) but is "not there yet," and that its detailed transmission mechanism sits with functions beyond the scope of a single communications-level interview. The contribution of MSME-municipality interaction to pathway change is therefore latent rather than realised: the institutional channel through which change could travel has been built, but the firm-tier uptake that would convert it into actual pathway change has not yet occurred.

6.2 Empirical and Theoretical Contributions

The empirical contribution is a detailed reading of how the post-flagship phase of European industrial policy is being lived in one mid-sized northern Swedish city. Umeå has built an unusually rich institutional architecture organised around two complementary capacity layers strategic brokerage from the Business Director's office and operational implementation through the Eco Industrial Park and project-funding pipelines within a regional innovation system the recent literature would classify as transformation-oriented rather than merely reorienting (Isaksen, Trippl & Mayer, 2022). The joint reading of all eight interviews shows that this architecture's effective reach into the manufacturing tier is uneven, that the binding institutional relationships of the firms run predominantly through national or EU scales, that the municipal "living circular hub" framing is only partially recognised even by the anchor firm closest to it, and that co-evolution unfolds along channels procurement, finance, and the anchor firm's binding conditions of infrastructure, electricity and competence that are not yet aligned with the logic the firms work within.

The theoretical contribution is the elaboration of co-evolution as a process described by five concepts direction, scale, channel, asset modification, and cultural register and the demonstration, through five phase-by-phase traces, that these concepts operate at different scales, directions and cadences for different firm–institution

pairs. The framework integrates EEG, RIS and sustainability transitions through Benner's (2026) cultural-toolkit perspective, relocated to the hinge between institutional and firm sides. Three theoretical implications follow. First, the cultural-toolkit perspective is a necessary rather than optional component of co-evolutionary analysis a point the anchor firm makes vivid, since it combines the closest municipal relationship in the corpus with explicit non-adoption of the municipal frame. Second, the open, multi-scalar conception of regional innovation systems is empirically vindicated, and the thesis adds the analytical category of latent fragmentation: a region whose formal supply-side architecture is dense but whose practical access conditions are uneven across the firm band. Third, the asymmetry of co-evolution operates on two axes between firms and institutions, and among firms in their access to institutions which sharpens Jolly et al.'s (2025) question of whose voice gets institutionalised.

6.3 Policy Implications

Three policy implications follow, each anchored in a theme. First, public procurement reform is the highest-leverage locally-tractable intervention: the convergence of Firm D and Firm E on the lowest-price-versus-lifecycle-cost problem (Theme 5) is striking, and translating existing lifecycle-cost tools into actual municipal procurement practice is a near-term intervention within the municipality's discretion. Second, SME export-finance reform is the highest-leverage national intervention: the SEK 200 million Swedish Export Credit threshold creates a size-based asymmetry that the EU's Clean Industrial Deal commitments to SMEs (European Commission, 2025a) will need to address if mid-sized cities are to retain export-oriented green firms. Third, the anchor firm's three binding conditions sustainable infrastructure, reliable green electricity, and specialised competence identify where municipal and regional capacity matters most for retaining large

green-manufacturing employment, and the firm's framing of these as a shared project ("we can't do it by our own, but we can solve it together") is an invitation to precisely the co-evolutionary coordination the Industriråd is positioned to convene.

6.4 Limitations and Future Research

Four limitations should be stated plainly. The study works from a small, deliberately chosen set of eight interviews enough to understand each case closely, but not a sample from which to generalise by counting, and resting on the firm side more heavily than on the two municipal voices that anchor it (Yin, 2018; Flyvbjerg, 2006). The anchor-firm conversation was held with the plant's communications director, so how sustainability requirements actually travel from the firm to its suppliers is something the study can confirm exists and is wanted, but cannot yet trace in detail. Timing is a third constraint: the European policy environment has kept shifting throughout the research, and the December 2025 Omnibus simplification of sustainability reporting (Council of the European Union, 2025) is only the most recent of the changes any continuation of this work will have to follow. And the study listens to firms and reads documents rather than living alongside them, so the cultural-toolkit perspective is read analytically rather than observed ethnographically.

Three directions for future research follow directly. The most valuable single next step is a follow-up with the anchor firm's logistics, purchasing or engineering functions, and with the northern suppliers the firm would like to draw into a denser regional flow, to evidence the supplier-tier transmission mechanism that this thesis has confirmed but only partially observed. A second direction is comparative:

placing Umeå alongside one or more structurally similar mid-sized industrial cities (Trondheim, Oulu, Tampere, or Linz as candidates) would test whether the latent-fragmentation category and the multi-scalar, asymmetric, culturally-mediated pattern of co-evolution generalise beyond the single case. A third direction is critical: the distributional questions Jolly et al. (2025) raise who benefits from the green path, and on whose terms invite closer attention to the wider northern-Swedish green-transition context, including the land-use, resource and Indigenous-rights dimensions flagged in section 4.1, which sit at the edge of this study's frame and which a fuller account of the region's industrial transformation would need to engage (Österlin & Raitio, 2020).

The case made here for the analytical centrality of MSMEs, anchor firms and mid-sized cities in Europe's post-flagship green industrial transformation rests, finally, on the patient mapping of one city, one institutional architecture, five manufacturing MSMEs and one anchor firm and on what their convergence and difference together reveal about a transition that will be carried, if it is carried at all, far more often in places like Umeå than in places like the flagship sites that have absorbed most of the attention. Co-evolution between firms and institutions in such places is multi-scalar, asymmetric, and culturally mediated; understanding it requires following the process across the scales at which it actually runs, rather than only at the scale at which institutional capacity is built.

References

Archibald, M. M., Ambagtsheer, R. C., Casey, M. G., & Lawless, M. (2019). Using Zoom videoconferencing for qualitative data collection: Perceptions and experiences of researchers and participants. *International Journal of Qualitative Methods,* 18, 1–8. https://doi.org/10.1177/1609406919874596

Agrawal, A., & Cockburn, I. (2003). The anchor tenant hypothesis: Exploring the role of large, local, R&D-intensive firms in regional innovation systems. *International Journal of Industrial Organization,* 21(9), 1227–1253. https://doi.org/10.1016/S0167-7187(03)00081-X

Asheim, B. T., & Gertler, M. S. (2005). The geography of innovation: Regional innovation systems. In J. Fagerberg, D. C. Mowery, & R. R. Nelson (Eds.), *The Oxford Handbook of Innovation* (pp. 291–317). Oxford University Press.

Benner, M. (2022). Retheorizing industrial–institutional coevolution: A multidimensional perspective. *Regional Studies,* 56(9), 1524–1537. https://doi.org/10.1080/00343404.2021.1949441

Benner, M. (2026). The coevolution of culture and industrial paths in regional restructuring and transition: A cultural toolkit approach. *Progress in Economic Geography*, 4(1), 100069. https://doi.org/10.1016/j.peg.2026.100069

Berger, R. (2015). Now I see it, now I don't: Researcher's position and reflexivity in qualitative research. *Qualitative Research,* 15(2), 219–234. https://doi.org/10.1177/1468794112468475

Bhaskar, R. (2008). *A realist theory of science* (3rd ed.). Routledge.

Borrás, S., & Edquist, C. (2013). The choice of innovation policy instruments. *Technological Forecasting and Social Change,* 80(8), 1513–1522. https://doi.org/10.1016/j.techfore.2013.03.002

Boschma, R., & Frenken, K. (2006). Why is economic geography not an evolutionary science? Towards an evolutionary economic geography. *Journal of Economic Geography,* 6(3), 273–302. https://doi.org/10.1093/jeg/lbi022

Boschma, R., & Martin, R. (Eds.). (2010). *The Handbook of Evolutionary Economic Geography.* Edward Elgar.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology,* 3(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Braun, V., & Clarke, V. (2022). *Thematic analysis*: A practical guide. SAGE.

Brinkmann, S., & Kvale, S. (2018). *Doing interviews* (2nd ed.). SAGE.

Chu, J., & Hassink, R. (2023). Advancing spatial ontology in evolutionary economic geography. *Cambridge Journal of Regions, Economy and Society,* 16(3), 391–404. https://doi.org/10.1093/cjres/rsad022

Coenen, L., Asheim, B., Bugge, M. M., & Herstad, S. J. (2017). Advancing regional innovation systems: What does evolutionary economic geography bring to the policy table? *Environment and Planning C: Politics and Space,* 35(4), 600–620. https://doi.org/10.1177/0263774X16646583

Coenen, L., Benneworth, P., & Truffer, B. (2012). Toward a spatial perspective on sustainability transitions. *Research Policy,* 41(6), 968–979. https://doi.org/10.1016/j.respol.2012.02.014

Cooke, P. (2001). Regional innovation systems, clusters, and the knowledge economy. *Industrial and Corporate Change,* 10(4), 945–974. https://doi.org/10.1093/icc/10.4.945

Council of the European Union. (2025, December 9). *Council and Parliament strike a deal to simplify sustainability reporting and due diligence requirements and boost EU competitiveness* [Press release].

David, P. A. (1985). Clio and the economics of QWERTY. *American Economic Review,* 75(2), 332–337.

Doloreux, D., & Parto, S. (2005). Regional innovation systems: Current discourse and unresolved issues. *Technology in Society,* 27(2), 133–153. https://doi.org/10.1016/j.techsoc.2005.01.002

Dubois, A., & Carson, D. (2016). Placing northern development: The case of North Sweden. *Local Economy,* 31(7), 795–811. https://doi.org/10.1177/0269094216674657

Essletzbichler, J. (2012). Evolutionary economic geography, institutions, and political economy. *Economic Geography,* 88(2), 187–194. https://doi.org/10.1111/j.1944-8287.2012.01147.x

Eurofound. (2025). *Battery manufacturing in the EU: From hope to crisis to hope again?* European Foundation for the Improvement of Living and Working Conditions.

European Commission. (2003). Commission Recommendation 2003/361/EC of 6 May 2003 concerning the definition of micro, small and medium-sized enterprises. *Official Journal of the European Union, L 124,* 36–41.

European Commission. (2020*). A new industrial strategy for Europe* (COM(2020) 102 final).

European Commission. (2023). 2023 *SME country fact sheet: Sweden.* SME Performance Review.

European Commission. (2024). 2024 *Country Report - Sweden* (SWD(2024) 627 final).

European Commission. (2025a). *The Clean Industrial Deal: A joint roadmap for competitiveness and decarbonisation* (COM(2025) 85 final).

Feldman, M. P. (2003). The locational dynamics of the US biotech industry: Knowledge externalities and the anchor hypothesis. *Industry and Innovation,* 10(3), 311–329. https://doi.org/10.1080/1366271032000141661

Fineman, J., & Vieweg, V. (2026). *Näringslivsanalys Västerbottens län 2019–2024.* Dun & Bradstreet Sverige.

Florence School of Regulation. (2024). *The Green Deal Industrial Plan.* European University Institute.

Flyvbjerg, B. (2006). Five misunderstandings about case-study research. *Qualitative Inquiry,* 12(2), 219–245. https://doi.org/10.1177/1077800405284363

Fortune. (2024, December 10). Europe's big battery ambitions are failing, and China is benefiting. *Fortune Europe.*

Frangenheim, A. (2023). Regional preconditions to shape interpath relations across regions: Two cases from the Austrian food sector. *European Planning Studies,* 31(2), 328–347. https://doi.org/10.1080/09654313.2022.2053661

Frenken, K., Van Oort, F., & Verburg, T. (2007). Related variety, unrelated variety and regional economic growth. *Regional Studies,* 41(5), 685–697. https://doi.org/10.1080/00343400601120296

Frisk, A., & Fineman, J. (2025). *Näringslivsanalys Umeå 2018–2023.* Dun & Bradstreet Sverige.

Galletti, A. (2013). The semi-structured interview as a repertoire of possibilities. In *The SAGE handbook of qualitative data collection.* SAGE.

Garud, R., & Karnøe, P. (2001). Path creation as a process of mindful deviation. In R. Garud & P. Karnøe (Eds.), *Path Dependence and Creation* (pp. 1–38). Lawrence Erlbaum.

Garud, R., Kumaraswamy, A., & Karnøe, P. (2010). Path dependence or path creation? *Journal of Management Studies, 47*(4), 760–774. https://doi.org/10.1111/j.1467-6486.2009.00914.x

Geels, F. W. (2002). Technological transitions as evolutionary reconfiguration processes: A multi-level perspective and a case-study. *Research Policy,* 31(8–9), 1257–1274. https://doi.org/10.1016/S0048-7333(02)00062-8

Geels, F. W. (2019). Socio-technical transitions to sustainability: A review of criticisms and elaborations of the multi-level perspective. *Current Opinion in Environmental Sustainability, 39,* 187–201. https://doi.org/10.1016/j.cosust.2019.06.009

Gong, H., & Hassink, R. (2019). Co-evolution in contemporary economic geography: Towards a theoretical framework. *Regional Studies, 53*(9), 1344–1355. https://doi.org/10.1080/00343404.2018.1494824

Government of Sweden. (2024). *Strategy for new industrialisation and societal transformation in Norrbotten and Västerbotten regions.* Regeringskansliet.

Grillitsch, M., & Sotarauta, M. (2020). Trinity of change agency, regional development paths and opportunity spaces. *Progress in Human Geography,* 44(4), 704–723. https://doi.org/10.1177/0309132519853870

Hodgson, G. M. (2006). What are institutions? *Journal of Economic Issues, 40*(1), 1–25. https://doi.org/10.1080/00213624.2006.11506879

Hug, M., Mayer, H., & Seidl, I. (2024). Transformative firm-level agency: A case study of small and medium-sized enterprises (SMEs) in the Swiss wood-processing industry. *Progress in Economic Geography, 2*(2), 100020. https://doi.org/10.1016/j.peg.2024.100020

Isaksen, A., & Trippl, M. (2017). Exogenously led and policy-supported new path development in peripheral regions. *Economic Geography, 93*(5), 436–457. https://doi.org/10.1080/00130095.2016.1276770

Isaksen, A., Trippl, M., & Mayer, H. (2022). Regional innovation systems in an era of grand societal challenges. *European Planning Studies, 30*(11), 2125–2138. https://doi.org/10.1080/09654313.2022.2084226

Jolly, S., Asheim, B., Benner, M., Calignano, G., Eadson, W., Gong, H., & Nilsen, T. (2025). Future-oriented green and just regional industrial path development: Towards a critical examination. *Progress in Economic Geography, 3*(2), 100049. https://doi.org/10.1016/j.peg.2025.100049

Karlsen, J., & Lund Pedersen, T. (2013). The role of anchor companies in thin regional innovation systems: Lessons from Norway. In B. T. Asheim & M. D. Parrilli (Eds.), *Interactive Learning for Innovation: A Key Driver within Clusters and Innovation Systems* (pp. 199–217). Palgrave Macmillan.

Karltorp, K., & Maltais, A. (2024). Financing green industrial transitions: A Swedish case study. *Energy and Climate Change, 5,* 100138. https://doi.org/10.1016/j.egycc.2024.100138

Kemp, R., Schot, J., & Hoogma, R. (1998). Regime shifts to sustainability through processes of niche formation: The approach of strategic niche management.

Technology Analysis & Strategic Management, 10(2), 175–198. https://doi.org/10.1080/09537329808524310

Kogler, D. F., Evenhuis, E., Giuliani, E., Martin, R., Uyarra, E., & Boschma, R. (2023). Re-imagining evolutionary economic geography. *Cambridge Journal of Regions, Economy and Society, 16*(3), 373–390. https://doi.org/10.1093/cjres/rsad029

Lincoln, Y. S., & Guba, E. G. (1985). *Naturalistic inquiry.* SAGE.

Madrid-Guijarro, A., & Duréndez, A. (2024). Sustainable development barriers and pressures in SMEs: The mediating effect of management commitment to environmental practices. *Business Strategy and the Environment, 33*(2), 949–967. https://doi.org/10.1002/bse.3537

Markard, J., Raven, R., & Truffer, B. (2012). Sustainability transitions: An emerging field of research and its prospects. *Research Policy, 41*(6), 955–967. https://doi.org/10.1016/j.respol.2012.02.013

Martin, R., & Sunley, P. (2006). Path dependence and regional economic evolution. *Journal of Economic Geography, 6*(4), 395–437. https://doi.org/10.1093/jeg/lbl012

Nordregio. (2024). *State of the Nordic Region 2024 - Chapter 6: Green transition of the labour market.* Nordregio.

Nilsson, M., Schubert, T., & Miörner, J. (2025). The productivity effects of regional anchors on local firms in Swedish regions between 2007 and 2019: Evidence from an expert-informed machine-learning approach. *Journal of Economic Geography, 25*(2), 215–234. https://doi.org/10.1093/jeg/lbae038

North, D. C. (1990). *Institutions, institutional change and economic performance.* Cambridge University Press.

OECD. (2023). *Climate policies and Sweden's green industrial revolution* (OECD Economics Department Working Papers No. 1778). OECD Publishing.

OECD. (2025a). *OECD environmental performance reviews: Sweden 2025.* OECD Publishing.

Österlin, C., & Raitio, K. (2020). Fragmented landscapes and planscapes - The double pressure of increasing natural resource exploitation on indigenous Sámi lands in northern Sweden. Resources, 9(9), 104. https://doi.org/10.3390/resources9090104

Patton, M. Q. (2015). *Qualitative research and evaluation methods* (4th ed.). SAGE.

Region Västerbotten. (2022). *Regional innovation strategy 2022–2030.* Region Västerbotten.

S&P Global Mobility. (2025, August). Why Europe is losing the gigafactory race to China. *S&P Global.*

Sayer, A. (2000). *Realism and social science.* SAGE.

Smith, A., Voß, J.-P., & Grin, J. (2010). Innovation studies and sustainability transitions: The allure of the multi-level perspective and its challenges. *Research Policy, 39*(4), 435–448. https://doi.org/10.1016/j.respol.2010.01.023

Sotarauta, M. (2018). Smart specialization and place leadership: Dreaming about shared visions, falling into policy traps? *Regional Studies, Regional Science, 5*(1), 190–203. https://doi.org/10.1080/21681376.2018.1480902

Stake, R. E. (1995). *The art of case study research.* SAGE.

Swedish Energy Agency. (2022, June 23). *Seven Swedish cities join the EU Mission for 100 climate neutral cities by 2030* [Press release]. Energimyndigheten.

Swidler, A. (1986). Culture in action: Symbols and strategies. *American Sociological Review,* 51(2), 273–286. https://doi.org/10.2307/2095521

Tagliapietra, S., & Trasi, C. (2024). *Northvolt's struggles: A cautionary tale for the EU Clean Industrial Deal* (Bruegel Analysis, 11 December). Bruegel.

Tödtling, F., & Trippl, M. (2005). One size fits all? Towards a differentiated regional innovation policy approach. *Research Policy, 34*(8), 1203–1219. https://doi.org/10.1016/j.respol.2005.01.018

Tracy, S. J. (2010). Qualitative quality: Eight "big-tent" criteria for excellent qualitative research. *Qualitative Inquiry, 16*(10), 837–851. https://doi.org/10.1177/1077800410383121

Trippl, M., Baumgartinger-Seiringer, S., Frangenheim, A., Isaksen, A., & Rypestøl, J. O. (2020). Unravelling green regional industrial path development: Regional preconditions, asset modification and agency. *Geoforum, 111,* 189–197. https://doi.org/10.1016/j.geoforum.2020.02.016

UBC Sustainable Cities Commission. (2023, April 12). Umeå's Climate Roadmap 2030 - *A pathway to follow.* Union of the Baltic Cities.

Umeå Municipality. (2024). *Umeå 2030 Climate Neutrality Action Plan.* NetZeroCities.

Vetenskapsrådet. (2017). *Good research practice.* Swedish Research Council.

Viable Cities. (2025). *Climate City Contract 2030 - 2025/2026 version.* Swedish Strategic Innovation Programme.

Yeung, H. W. (2019). Rethinking mechanism and process in the geographical analysis of uneven development. *Dialogues in Human Geography, 9*(3), 226–255. https://doi.org/10.1177/2043820619870193

Yin, R. K. (2018). Case study research and applications: *Design and methods* (6th ed.). SAGE.

Yunusbayeva, G., Pohl, H., Hjelmberg, J., & Bhatti, W. (2025). Balancing barriers and enablers: Swedish SMEs' journey toward contributing to SDGs. *Business Strategy and the Environment, 34*(8), 9605–9619. https://doi.org/10.1002/bse.70089